

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250269276

Kind Code

A1

Publication Date

August 28, 2025

Inventor(s)

Gaither; Geoffrey David et al.

CONTENT MANAGEMENT DURING VEHICLE CHARGING

Abstract

An example operation includes one or more of determining a remaining time until charging of a vehicle is complete and transmitting content to the vehicle based on the remaining time.

Inventors: Gaither; Geoffrey David (Ann Arbor, MI), Swartz; James G. (Prosper, TX), Wilder; James D. (Dallas, TX)

Applicant: TOYOTA MOTOR NORTH AMERICA, INC. (Plano, TX)

Family ID: 1000007715487

Assignee: TOYOTA MOTOR NORTH AMERICA, INC. (Plano, TX); TOYOTA JIDOSHA KABUSHIKI KAISHA (AICHI-KEN, JP)

Appl. No.: 18/589360

Filed: February 27, 2024

Publication Classification

Int. Cl.: A63F13/355 (20140101); B60L53/66 (20190101); H04W4/40 (20180101)

U.S. Cl.:

CPC A63F13/355 (20140902); B60L53/665 (20190201); H04W4/40 (20180201);

Background/Summary

BACKGROUND

[0001] Vehicles or transports, such as cars, motorcycles, trucks, planes, trains, etc., generally provide transportation needs to occupants and/or goods in a variety of ways. Functions related to vehicles may be identified and utilized by various computing devices, such as a smartphone or a

computer located on and/or off the vehicle.

SUMMARY

[0002] One example embodiment provides a method that includes one or more of determining a remaining time until charging of a vehicle is complete, and transmitting content to the vehicle based on the remaining time.

[0003] Another example embodiment may include a system that includes a memory and a processor configured to determine a remaining time until charging of a vehicle is complete, and transmit content to the vehicle based on the remaining time.

[0004] Another example embodiment may include a non-transitory computer readable medium configured to store instructions that when executed cause a processor to perform determining a remaining time until charging of a vehicle is complete, and transmitting content to the vehicle based on the remaining time.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1A illustrates an example system diagram of a vehicle charging at a charging station and the receiving content from a server, according to example embodiments.

[0006] FIG. 1B illustrates a network diagram of a vehicle using a charging station and receiving content and performing vehicle operations based on the content, according to example embodiments.

[0007] FIG. 2A illustrates a vehicle network diagram, according to example embodiments.

[0008] FIG. 2B illustrates another vehicle network diagram, according to example embodiments.

[0009] FIG. 2C illustrates yet another vehicle network diagram, according to example embodiments.

[0010] FIG. 2D illustrates a further vehicle network diagram, according to example embodiments.

[0011] FIG. 2E illustrates a flow diagram, according to example embodiments.

[0012] FIG. 2F illustrates another flow diagram, according to example embodiments.

[0013] FIG. 3A illustrates an Artificial Intelligence (AI)/Machine Learning (ML) network diagram for integrating an artificial intelligence (AI) model into any decision point in the example embodiments.

[0014] FIG. 3B illustrates a process for developing an Artificial Intelligence (AI)/Machine Learning (ML) model that supports AI-assisted vehicle or occupant decision points.

[0015] FIG. 3C illustrates a process for utilizing an Artificial Intelligence (AI)/Machine Learning (ML) model that supports AI-assisted vehicle or occupant decision points.

[0016] FIG. 3D illustrates a machine learning network diagram, according to example embodiments.

[0017] FIG. 3E illustrates a machine learning network diagram, according to example embodiments.

[0018] FIG. 4A illustrates a diagram depicting electrification of one or more elements, according to example embodiments.

[0019] FIG. 4B illustrates a diagram depicting interconnections between different elements, according to example embodiments.

[0020] FIG. 4C illustrates a further diagram depicting interconnections between different elements, according to example embodiments.

[0021] FIG. 4D illustrates yet a further diagram depicting interconnections between elements, according to example embodiments.

[0022] FIG. 4E illustrates yet a further diagram depicting an example of vehicles performing secured Vehicle-to-Vehicle (V2V) communications using security certificates, according to

example embodiments.

[0023] FIG. 5A illustrates an example vehicle configuration for managing database transactions associated with a vehicle, according to example embodiments.

[0024] FIG. 5B illustrates an example blockchain group, according to example embodiments.

[0025] FIG. 5C illustrates an example interaction between elements and a blockchain, according to example embodiments.

[0026] FIG. 5D illustrates an example data block interaction, according to example embodiments.

[0027] FIG. 5E illustrates a blockchain network diagram, according to example embodiments.

[0028] FIG. 5F illustrates an example new data block, according to example embodiments.

[0029] FIG. 6 illustrates an example system that supports one or more of the example embodiments.

DETAILED DESCRIPTION

[0030] It will be readily understood that the instant components, as generally described and illustrated in the figures herein, may be arranged and designed in a wide variety of different configurations. Thus, the following detailed description of the embodiments of at least one of a method, apparatus, computer readable storage medium and system, as represented in the attached figures, is not intended to limit the scope of the application as claimed but is merely representative of selected embodiments. Multiple embodiments depicted herein are not intended to limit the scope of the solution. The computer-readable storage medium may be a non-transitory computer readable medium or a non-transitory computer readable storage medium.

[0031] Communications between the vehicle(s) and certain entities, such as remote servers, other vehicles, and local computing devices (e.g., smartphones, personal computers, vehicle-embedded computers, etc.) may be sent and/or received and processed by one or more ‘components’ which may be hardware, firmware, software, or a combination thereof. The components may be part of any of these entities or computing devices or certain other computing devices. In one example, consensus decisions related to blockchain transactions may be performed by one or more computing devices or components (which may be any element described and/or depicted herein) associated with the vehicle(s) and one or more of the components outside or at a remote location from the vehicle(s).

[0032] The instant features, structures, or characteristics described in this specification may be combined in any suitable manner in one or more embodiments. For example, the usage of the phrases “example embodiments,” “some embodiments,” “a first embodiment”, or other similar language throughout this specification refers to the fact that a particular feature, structure, or characteristic described in connection with the one or more embodiments may be included in one or more other embodiments described or depicted herein. Thus, the one or more embodiments, described or depicted throughout this specification can all refer to the same embodiment. Thus, these embodiments may work in conjunction with any of the other embodiments, may not be functionally separate, and the described features, structures, or characteristics may be combined in any suitable manner in one or more embodiments. Although described in a particular manner, by example only, or more feature(s), element(s), and step(s) described herein may be utilized together and in various combinations, without exclusivity, unless expressly indicated otherwise herein. In the figures, any connection between elements can permit one-way and/or two-way communication, even if the depicted connection is a one-way or two-way connection, such as an arrow.

[0033] In the instant solution, a vehicle may include one or more of cars, trucks, Internal Combustion Engine (ICE) vehicles, battery electric vehicle (BEV), fuel cell vehicles, any vehicle utilizing renewable sources, hybrid vehicles, e-Pallettes, buses, motorcycles, scooters, bicycles, boats, recreational vehicles, planes, drones, Unmanned Aerial Vehicle (UAV) and any object that may be used to transport people and/or goods from one location to another.

[0034] In addition, while the term “message” may have been used in the description of embodiments, other types of network data, such as, a packet, frame, datagram, etc. may also be

used. Furthermore, while certain types of messages and signaling may be depicted in exemplary embodiments they are not limited to a certain type of message and signaling.

[0035] Example embodiments provide methods, systems, components, non-transitory computer readable medium, devices, and/or networks, which provide at least one of a transport (also referred to as a vehicle or car herein), a data collection system, a data monitoring system, a verification system, an authorization system, and a vehicle data distribution system. The vehicle status condition data received in the form of communication messages, such as wireless data network communications and/or wired communication messages, may be processed to identify vehicle status conditions and provide feedback on the condition and/or changes of a vehicle. In one example, a user profile may be applied to a particular vehicle to authorize a current vehicle event, service stops at service stations, to authorize subsequent vehicle rental services, and enable vehicle-to-vehicle communications.

[0036] Within the communication infrastructure, a decentralized database is a distributed storage system which includes multiple nodes that communicate with each other. A blockchain is an example of a decentralized database, which includes an append-only immutable data structure (i.e., a distributed ledger) capable of maintaining records between untrusted parties. The untrusted parties are referred to herein as peers, nodes, or peer nodes. Each peer maintains a copy of the database records, and no single peer can modify the database records without a consensus being reached among the distributed peers. For example, the peers may execute a consensus protocol to validate blockchain storage entries, group the storage entries into blocks, and build a hash chain via the blocks. This process forms the ledger by ordering the storage entries, as is necessary, for consistency. In public or permissionless blockchains, anyone can participate without a specific identity. Public blockchains can involve crypto-currencies and use consensus-based on various protocols such as proof of work (PoW). Conversely, a permissioned blockchain database can secure interactions among a group of entities, which share a common goal, but which do not or cannot fully trust one another, such as businesses that exchange funds, goods, information, and the like. The instant solution can function in a permissioned and/or a permissionless blockchain setting.

[0037] Smart contracts are trusted distributed applications which leverage tamper-proof properties of the shared or distributed ledger (which may be in the form of a blockchain) and an underlying agreement between member nodes, which is referred to as an endorsement or endorsement policy. In general, blockchain entries are “endorsed” before being committed to the blockchain while entries which are not endorsed are disregarded. A typical endorsement policy allows smart contract executable code to specify endorsers for an entry in the form of a set of peer nodes that are necessary for endorsement. When a client sends the entry to the peers specified in the endorsement policy, the entry is executed to validate the entry. After validation, the entries enter an ordering phase in which a consensus protocol produces an ordered sequence of endorsed entries grouped into blocks.

[0038] Nodes are the communication entities of the blockchain system. A “node” may perform a logical function in the sense that multiple nodes of different types can run on the same physical server. Nodes are grouped in trust domains and are associated with logical entities that control them in various ways. Nodes may include different types, such as a client or submitting-client node, which submits an entry-invocation to an endorser (e.g., peer), and broadcasts entry proposals to an ordering service (e.g., ordering node). Another type of node is a peer node, which can receive client submitted entries, commit the entries, and maintain a state and a copy of the ledger of blockchain entries. Peers can also have the role of an endorser. An ordering-service-node or orderer is a node running the communication service for all nodes and which implements a delivery guarantee, such as a broadcast to each of the peer nodes in the system when committing entries and modifying a world state of the blockchain. The world state can constitute the initial blockchain entry, which normally includes control and setup information.

[0039] A ledger is a sequenced, tamper-resistant record of all state transitions of a blockchain. State

transitions may result from smart contract executable code invocations (i.e., entries) submitted by participating parties (e.g., client nodes, ordering nodes, endorser nodes, peer nodes, etc.). An entry may result in a set of asset key-value pairs being committed to the ledger as one or more operands, such as creates, updates, deletes, and the like. The ledger includes a blockchain (also referred to as a chain), which stores an immutable, sequenced record in blocks. The ledger also includes a state database, which maintains a current state of the blockchain. There is typically one ledger per channel. Each peer node maintains a copy of the ledger for each channel of which they are a member.

[0040] A chain is an entry log structured as hash-linked blocks, and each block contains a sequence of N entries where N is equal to or greater than one. The block header includes a hash of the blocks' entries, as well as a hash of the prior block's header. In this way, all entries on the ledger may be sequenced and cryptographically linked together. Accordingly, it is not possible to tamper with the ledger data without breaking the hash links. A hash of a most recently added blockchain block represents every entry on the chain that has come before it, making it possible to ensure that all peer nodes are in a consistent and trusted state. The chain may be stored on a peer node file system (i.e., local, attached storage, cloud, etc.), efficiently supporting the append-only nature of the blockchain workload.

[0041] The current state of the immutable ledger represents the latest values for all keys that are included in the chain entry log. Since the current state represents the latest key values known to a channel, it is sometimes referred to as a world state. Smart contract executable code invocations execute entries against the current state data of the ledger. To make these smart contract executable code interactions efficient, the latest values of the keys may be stored in a state database. The state database may be simply an indexed view into the chain's entry log and can therefore be regenerated from the chain at any time. The state database may automatically be recovered (or generated if needed) upon peer node startup and before entries are accepted.

[0042] A blockchain is different from a traditional database in that the blockchain is not a central storage but rather a decentralized, immutable, and secure storage, where nodes must share in changes to records in the storage. Some properties that are inherent in blockchain and which help implement the blockchain include, but are not limited to, an immutable ledger, smart contracts, security, privacy, decentralization, consensus, endorsement, accessibility, and the like.

[0043] Example embodiments provide a service to a particular vehicle and/or a user profile that is applied to the vehicle. For example, a user may be the owner of a vehicle or the operator of a vehicle owned by another party. The vehicle may require service at certain intervals, and the service needs may require authorization before permitting the services to be received. Also, service centers may offer services to vehicles in a nearby area based on the vehicle's current route plan and a relative level of service requirements (e.g., immediate, severe, intermediate, minor, etc.). The vehicle needs may be monitored via one or more vehicle and/or road sensors or cameras, which report sensed data to a central controller computer device in and/or apart from the vehicle. This data is forwarded to a management server for review and action. A sensor may be located on one or more of the interior of the vehicle, the exterior of the vehicle, on a fixed object apart from the vehicle, and on another vehicle proximate the vehicle. The sensor may also be associated with the vehicle's speed, the vehicle's braking, the vehicle's acceleration, fuel levels, service needs, the gear-shifting of the vehicle, the vehicle's steering, and the like. A sensor, as described herein, may also be a device, such as a wireless device in and/or proximate to the vehicle. Also, sensor information may be used to identify whether the vehicle is operating safely and whether an occupant has engaged in any unexpected vehicle conditions, such as during a vehicle access and/or utilization period. Vehicle information collected before, during and/or after a vehicle's operation may be identified and stored in a transaction on a shared/distributed ledger, which may be generated and committed to the immutable ledger as determined by a permission granting consortium, and thus in a "decentralized" manner, such as via a blockchain membership group.

[0044] Each interested party (i.e., owner, user, company, agency, etc.) may want to limit the exposure of private information, and therefore the blockchain and its immutability can be used to manage permissions for each particular user vehicle profile. A smart contract may be used to provide compensation, quantify a user profile score/rating/review, apply vehicle event permissions, determine when service is needed, identify a collision and/or degradation event, identify a safety concern event, identify parties to the event and provide distribution to registered entities seeking access to such vehicle event data. Also, the results may be identified, and the necessary information can be shared among the registered companies and/or individuals based on a consensus approach associated with the blockchain. Such an approach may not be implemented on a traditional centralized database.

[0045] Various driving systems of the instant solution can utilize software, an array of sensors as well as machine learning functionality, light detection and ranging (LiDAR) projectors, radar, ultrasonic sensors, etc. to create a map of terrain and road that a vehicle can use for navigation and other purposes. In some embodiments, global positioning system (GPS), maps, cameras, sensors, and the like can also be used in autonomous vehicles in place of LiDAR.

[0046] The instant solution includes, in certain embodiments, authorizing a vehicle for service via an automated and quick authentication scheme. For example, driving up to a charging station or fuel pump may be performed by a vehicle operator or an autonomous vehicle and the authorization to receive charge or fuel may be performed without any delays provided the authorization is received by the service and/or charging station. A vehicle may provide a communication signal that provides an identification of a vehicle that has a currently active profile linked to an account that is authorized to accept a service, which can be later rectified by compensation. Additional measures may be used to provide further authentication, such as another identifier may be sent from the user's device wirelessly to the service center to replace or supplement the first authorization effort between the vehicle and the service center with an additional authorization effort.

[0047] Data shared and received may be stored in a database, which maintains data in one single database (e.g., database server) and generally at one particular location. This location is often a central computer, for example, a desktop central processing unit (CPU), a server CPU, or a mainframe computer. Information stored on a centralized database is typically accessible from multiple different points. A centralized database is easy to manage, maintain, and control, especially for purposes of security because of its single location. Within a centralized database, data redundancy is minimized as a single storing place of all data also implies that a given set of data only has one primary record. A blockchain may be used for storing vehicle-related data and transactions.

[0048] Any of the actions described herein may be performed by one or more processors (such as a microprocessor, a sensor, an Electronic Control Unit (ECU), a head unit, and the like), with or without memory, which may be located on-board the vehicle and/or off-board the vehicle (such as a server, computer, mobile/wireless device, etc.). The one or more processors may communicate with other memory and/or other processors on-board or off-board other vehicles to utilize data being sent by and/or to the vehicle. The one or more processors and the other processors can send data, receive data, and utilize this data to perform one or more of the actions described or depicted herein.

[0049] FIG. 1A illustrates an example system diagram of a vehicle charging at a charging station and the process of receiving content managed by a server, according to example embodiments. Referring to FIG. 1A, the system configuration **100** includes a vehicle **130** being identified as a candidate to initiate charging at a charging station **152** or charging point, hereinafter used interchangeably. A server **160** may be a remote computer in a cloud network or a local computer operating near the vehicle **130**, such as a mobile device or other computing device. Once a vehicle **130** is identified as a candidate for power charging (e.g., electrical charging), the server **160** may assign the vehicle to a particular charging station **152** and the vehicle **130** may begin receiving

charge **112**. The vehicle **130** may then communicate with the charging station **152** to initiate the charging process **114**. An initial vehicle charge level may quickly change to an elevated/higher/greater amount of charge which is continuously monitored by the server **160**. The amount of time, the updated charge level, at any particular time, and other charge attributes may be monitored by the server **160** in an ongoing basis to determine a status of the vehicle **130**. A prediction may be made by the server **160** based on known data as to how long the vehicle should charge or will need to charge to reach an optimal charge level (e.g., 60 percent, 70 percent, 80 percent or higher). The estimated charge time may be determined **116** based on a type of vehicle, the type of charging station, previous charge durations on a per vehicle basis, etc. All such information may be used as the basis for predicting how long the vehicle **130** should remain charging at the charging station **152**.

[0050] In the event that the vehicle **130** has a predefined amount of time to charge, such as a time period that can be identified as an estimated charging window, content may be transmitted **118** to the vehicle from a server **160** or other content management entity. The content may be a movie, a commercial, a television show, video, audio, etc. The content may also be a game, such as an online game that a user may play by a controller or other actions provided as feedback to manipulate the game session. The server **160** may identify and maintain a list of actions **122** performed by the occupant of the vehicle **130**. For example, the occupant may be playing a game and using a controller, such as a mobile device and/or a computing device integrated with the vehicle, performing detectable gestures by moving their hands, etc. As the game is played in the vehicle **130**, the server **160** may control one or more functions and parts of the vehicle **130** based on the outcome of the game actions.

[0051] Continuing with the same example, as an occupant is playing a game based on content sent to the vehicle computing device and/or the occupant's mobile device, one or more vehicle functions may be enabled **124** based on actions, such as a signal transmitted from the server **160** and/or enabled by the control signals provided by the computing device and/or the vehicle **130**. In one example, if an occupant uses an avatar in a game to jump off a bridge, the vehicle **130** may enable air to blow from the vent system and/or shake controllable portions of the vehicle to simulate a free fall experience. As the action is performed by the occupant on the in-vehicle controller, the server **160** may receive the action, such as a digitized command, and respond by transmitting **126** a vehicle control function. Another example may include scented air being released by a diffuser installed in the vehicle based on a select game action. The seat in the vehicle may move to simulate a change in position or status based on a game action performed. Any controllable aspect of the vehicle may be modified to simulate the game action identified.

[0052] While a vehicle **130** is charging, certain third-party affiliations may be identified from a user and/or vehicle profile associated with the vehicle **130**. Examples may include subscription services, a current status of purchased orders (e.g., food delivery orders, purchases from merchants, etc.) which require delivery, all of which may be identified to consider routing those orders to the vehicle during charging. In one example, the charging station **152** may be in a busier part of town than the user's home address, and it may be prudent to consider this temporary location as a delivery location to reduce the number of miles the delivery vehicle may normally have to drive to reach the user's home. The affiliations identified by the server **160** may yield a list of locations and/or services which are included in a user's profile as pending services, such as pending deliveries, pending subscriptions, etc. Any one of the affiliations with a pending status may be dispatched to include an order or service during the estimated time window for charging the vehicle. Once an affiliation is identified, an item or service may be selected to provide to the vehicle. This may include one, two, or more items, which can include physical items, such as meal delivery, parcel delivery and non-tangible items, such as streaming movies, music, etc. A user profile may have a preference setup so that only computer-based content is sent to the vehicle instead of deliveries, or that all possible deliveries are sent to a charging station at any given time

for that particular vehicle.

[0053] FIG. 1B illustrates a network diagram of a vehicle using a charging station and a content being sent to the vehicle during the charging session, according to example embodiments.

Referring to FIG. 1B, the network **150** includes a vehicle management server **160** which may maintain a status of the vehicle **130** which is having a charge cycle at a charging point or charging station **152**. The currently charging vehicle **130** may have an initial charge level, such as 10 percent or a low amount of charge respective to a full charge level of 100 percent. Any vehicle that is attempting to charge below an initial minimum charge level, such as 30 percent, may be charged more per unit of charge received since the vehicle may not be considered charge deficient at that particular time since it is not over the minimum charge level. The amount of charge required may be used as a basis for time to charge and to select content to send to the vehicle and share with the occupant of the vehicle.

[0054] In this example, the vehicle **130** may begin charging at the charging station **152** and an estimated charge time **162** may be determined by the vehicle management server **160**, or another computing device, to identify whether the time remaining until the vehicle **130** exits the charging station **152** is a long enough period of time for the vehicle **130** to receive content from the server **160** and consume the provided content **164** in a time period that is less than or close to the estimated charge time **162**. The moment the vehicle **130** is linked with the charging station **152**, the timer may begin to measure a total time estimated for the vehicle **130** to complete the charge as a function of the current time, an amount of charge the vehicle has currently, and a rate of charge of the charging station **152**. A content selection, such as a movie, a video game, etc., may have a corresponding use time associated with the selection, and the decision made by the vehicle management server **160** to dispatch the content to the vehicle **130** and its corresponding computer system, such as monitor and computer integrated with the vehicle and/or an occupant's mobile device inside the vehicle, may be based on the estimated use time. When the amount of time is less than or near the predicted time remaining, that particular selection may be dispatched for use by the occupant.

[0055] In one example, the system application may utilize the estimated charging time for delivering content to the vehicle and/or occupant(s). Recognizing that vehicle owners may charge 80% or more of their battery capacity which can take a significant amount of time, the system application capitalizes on this available time duration by providing the occupant(s) with entertainment and/or informational data. Examples may include streaming media, news updates, games and/or interactive content tailored to the predicted duration of the charging session. The latest news may also be delivered via video, audio, and/or text, depending on the user's preferences and the content may be premium and licensed to the charge service. Also, games, quizzes, educational content, or even guided meditation sessions may be offered to engage the users interactively. The system application may provide a selection of the content that users can consume during the time needed to charge their vehicle. For instance, if a charging session is expected to take 30 minutes, the system application may suggest content that can be completed in that identified time duration, so the user has a satisfying start-to-finish experience at the station. The content delivery may be automated and personalized based on known occupant preferences stored in an online profile for the vehicle and/or the occupant's mobile device as soon as the charging session begins. Over time, the system application responsible for sharing content may learn the user's content preferences and suggest personalized options for each charging session which are different from a previous session. The content offerings may also be based on the time of day, location, and/or charging speed, and may be dynamically adjusted so the options are different if the charging time is shorter or longer than initially expected.

[0056] In one example, the system application may coordinate certain services, such as vehicle washes and repairs while parked at the charging station. The scheduling of these services may be based on the expected duration of the vehicle's stay, communicated by the system application's

ability to predict how long the vehicle will be charging (e.g., 38 minutes from 20% to 80% battery capacity). The prediction may assist with scheduling of certain ancillary services received within that window. Using a combination of predictive analytics to estimate charging times and a scheduling application for creating and confirming service appointments, the vehicle owner may be able to optimize the use of their time while charging.

[0057] In one example, the system application uses predictive capabilities for determining the vehicle's accessibility window for various services or goods being delivered. The system application operating on the server **160** can predict the available time by assessing factors such as the vehicle's current state, required maintenance, such as fluid levels, tire pressure, or headlight functionality, etc., and the user's consumption of goods, such as videos watched recently, recent food purchases, etc. The application can then coordinate with service providers to secure appointments for those items within the timeframe allotted, and by presenting users with options and associated costs for services.

[0058] While in a charging status, the user may be able to leave the vehicle and participate in a nearby activity, such as entertainment or dining. An autonomous vehicle, such as a taxi, car, scooter, or other transportation item may be dispatched to visit the vehicle **130** and collect the user and take them on a tour of the area or to a nearby restaurant during the charging period. The vehicle may deliver food or other items prior to having the user enter the autonomous vehicle. The schedule of time and activities may be performed autonomously based on the predicted charging time window of the vehicle.

[0059] The vehicle seats, such as the driver seat or 'cockpit', may be well-suited for gaming, with ergonomic seating designed for good visibility and enhanced video displays, and audio surround systems are standard in features in many new electrical vehicles (EVs). EVs may soon become the ideal game console with tactile and sensory effects delivered through vehicle features, such as massage seating and directional air conditioning, which can be paired to what is going on in the game environment. For example, warm or cold air may stream from the vehicle ventilation nozzles depending on a game situation (e.g., the north pole, the jungle, etc.). Also, an on-screen explosion may be intensified by sudden activation of the heated seats, the movement or vibration of the seat, etc. The seat belt tensioner may tighten to intensify a quick or abrupt movement if a game activity includes a sudden stop or an impact. Additionally, vehicle scenting may infuse the vehicle's interior with appropriate aromas, such as pine, fresh grass, fruits, smoke, when the game activity includes entering a new environment.

[0060] In one example, an occupant may be prevented from starting another game after one is finished if there is not enough estimated time remaining for a satisfactory gaming session to occur before the vehicle's charging is completed. This feature is based on real-time data and helps to ensure that the occupant's game session is not interrupted abruptly when the charging is complete yet the game session is not complete. The application may permit occupants to exit a game or stop the game when their vehicle charging is complete. In one example, a video recording of the game played is sent to a user device associated with that user after exiting from the game. Also, a pending or active game can be saved and sent to another device to be finished at a later time.

[0061] In one example, vehicle hardware sensors in the vehicle may monitor the charging progress. The sensors can measure the current charging rate, battery status, and remaining time until a full charge is obtained. Communication between the vehicle's charging and gaming systems may be established. The system application may collect and analyze statistical data related to the gaming session(s). The data can include game duration, user performance, and preferences. In one embodiment, artificial intelligence (AI) models and machine learning functionality are utilized to predict the optimal timing for the next gaming session based on historical use time data and current charging status. The predictions may include selecting a type of game that is appropriate for a type of vehicle based on actions in the game and control functions of the vehicle to provide a simulation environment that is ideal for the type of game being played. Another prediction may be a likely

start and finish timeline for a game that is closest to the estimated charge time of the vehicle so as not to have the occupant be interrupted during a game play session due to a charge session ending and so as not to have the game duration be too short when the vehicle will have to continue charging for an appreciable amount of time after the game is finished. The system application may provide notifications and/or alerts during the game play session when it appears there will not be enough time to finish the game since the charging is completed prior to the game session. The notifications may be sent by the server **160** to the vehicle **130** and a corresponding computing device and/or the occupant's personal device(s). The notifications may appear as part of the game or via a separate notification platform.

[0062] In one example, an interface may be designed within the vehicle's entertainment system or via a connected mobile application to permit users to exit games or stop games when charging is complete. Video recording functionality is implemented within the gaming system application, to capture the user's gameplay. This recorded video can be stored temporarily or uploaded to a cloud server for later access. Recorded gameplay videos may be sent to a user's associated device, such as a smartphone or tablet. Certain examples may include a mobile application that users can install on their smartphones and/or tablets. The application permits users to interact with the gaming system, receive notifications, and access recorded gameplay videos. The gaming system may be integrated into the vehicle's entertainment system (such as in the head unit of the vehicle), providing a seamless experience for users within the vehicle.

[0063] One example may include a process that includes determining a remaining time until charging of a vehicle is complete and transmitting content to the vehicle based on the remaining time. The time span may be identified based on an estimated calculation that is performed according to a type of vehicle, a type of charging station, an amount of current charge of the vehicle, an amount of charge needed and/or a current demand for charging station use by various other EVs in the area.

[0064] The process may also include identifying an application associated with a profile of an occupant of the vehicle and initiating the application while the vehicle is charging when the remaining time is within a time duration to use the application. The application may be a type of data service, such as streaming videos, movies, news, etc., and/or a type of other entertainment, such as a game or related interactive experience (e.g., trivia, online multi-user activities, etc.). The user profile associated with the vehicle and/or a user device of the vehicle occupant(s) may provide information about previous content use, current interests, age, occupation, interests, etc., any of which can be used to select a video, audio, or game to share with the occupant during a charging session. The process may also include identifying one or more actions performed by an occupant while using an application and enabling one or more vehicle functions responsive to the one or more actions identified in the application. The actions may be selections made in an interactive multimedia session, such as a game or other selection-based activity. For example, an action to jump or move quickly may increase an air ventilation function of the vehicle to blow air at a faster rate. Another example may be an action to punch or kick an opponent may cause a vibration or movement in the seat control of the vehicle. The one or more vehicle functions may include a chair movement, a change in air ventilation speed or temperature, a change in light control, and a change in an air infuser. The content transmitted may be streaming content of a video and/or game associated with an occupant of the vehicle. The process may also include determining a game session associated with the application has ended prematurely at an updated remaining time and determining not to initiate another game session based on the updated remaining time. In one example, if a game is expected to average 7 minutes and the charging time is expected to be completed in less than 3 minutes, the decision to permit another game may be stopped by the system application. In such a scenario, a short content session may be initiated, such as recent news headlines and other types of content instead of the game in an attempt to fill the remainder of the charging timeline after the game has ended. The process may also include determining the charging

is complete, determining a game session associated with the application is not completed, and forwarding game session data to a user device. The occupant can then access the game at a later time as the game profile may be stored in a cloud server application so the user can resume the game on their personal device after leaving the vehicle.

[0065] In one embodiment, the system offers interactive lessons on environmental conservation, renewable energy, and sustainable living during a vehicle's charging time. Users can learn about solar power, wind energy, recycling, and wildlife preservation. Quizzes and mini-games keep passengers engaged. The system educates and inspires passengers about environmental sustainability, renewable energy, and ecological awareness. When the vehicle is plugged in for charging, passengers can activate the ecological education mode through the infotainment system or a dedicated button. The vehicle's screens become educational interfaces, whether integrated into the dashboard, headrests, or backseat displays. Content is delivered through videos, interactive modules, quizzes, and captivating visuals. The videos are arranged in classroom curriculum format, allowing occupants to participate as pseudo-students and progress through the curriculum with each charging session. Occupants can learn various topics, including how solar panels work, harnessing wind power, understanding hydroelectric dams, battery chemistry, the importance of recycling, and wildlife conservation. The system also contains interactive elements, including quizzes on environmental topics, virtual tours of ecology-related locations, and ecology-themed games. As passengers engage with the content, they earn points based on participation. These points can be converted into charging credits or donated to environmental organizations. The system can be customized for individual learning experiences. For instance, the system can serve simplified content for younger passengers and in-depth information for adults, enthusiasts, or occupants who have progressed through the quizzes and scored high points.

[0066] In one embodiment, the instant solution encourages physical activity while a vehicle charges. During charging sessions, the vehicle's infotainment system transforms into a fitness hub. Passengers can engage in various exercises without leaving the vehicle. The vehicle synchronizes with compatible exercise machines (e.g., treadmills, ellipticals, rowers) available at charging stations. Real-time heart rate data is displayed on the vehicle screens, allowing users to monitor their workout intensity. Passengers choose from various exercise modes, including cardio, strength training, and flexibility routines, and set their workout pace and mood through music selection. The system tracks the time spent in target heart rate zones during exercise, and passengers receive feedback on their effort level and overall fitness intensity. As users engage in workouts, they earn charging credits or other incentives. The system can be tailored to individuals' fitness experiences with gentle exercises for beginners and high-intensity workouts for advanced users.

[0067] In one embodiment, the system utilizes a vehicle charging session to engage occupants in a collaborative art session. As the vehicle charges, its infotainment system becomes a digital canvas. Passengers, whether friends, family, or strangers, contribute to an evolving artwork. Passengers can sketch, doodle, or create intricate designs with touchscreens or stylus pens. Collaborators can add poetry, short stories, or snippets of their thoughts. Passengers can create new music by utilizing the vehicle's audio system to play music, display lyrics, and invite other passengers to sing or compose additional verses. Each charging session adds content to existing artwork, whether new chapters in a novel, different graphics in a piece of art, or new verses in a song. The vehicle archives past creations, creating a gallery accessible during future rides. Users can revisit their collaborative masterpieces or explore others' contributions. The ad-hoc nature of creating art while in the vehicle allows for collaboration and creativity beyond the normal setting of a studio or home setting.

[0068] In one embodiment, the system provides guided virtual tours while a vehicle charges. When the vehicle is plugged in for charging, the infotainment system activates the virtual tour guide mode. Passengers can choose from a menu of destinations or allow the system to make the selection. The vehicle screens display detailed three-dimensional (3D) models of nearby attractions, allowing users to virtually explore historical buildings, national parks, or iconic

landmarks. Short videos provide context, historical anecdotes, and fascinating facts. Users can enable narration and select from optional narrative voices as they embark on a tour. During a tour, users can select interactive points of interest within the 3D models to learn about architectural styles, flora and fauna, or geological formations. The system also tests knowledge from the tour through quizzes, with correct answers going toward charging credits. For vehicles equipped with augmented reality (AR) capabilities, passengers can overlay historical images onto the current landscape to witness how areas appeared at different historical points.

[0069] In one embodiment, the system transforms the vehicle's interior into a productive workspace during charging. Occupants can access tailored productivity applications and tools seamlessly integrated into the vehicle's infotainment system. The tools include popular business software such as word processors, spreadsheets, and presentation software, enabling users to work on documents, presentations, and spreadsheets while waiting for the vehicle to charge. Additionally, collaborative tools like video conferencing software, project management platforms, and shared document editors facilitate real-time collaboration with colleagues or clients, allowing occupants to conduct virtual meetings, brainstorming sessions, or project reviews from the comfort of their vehicle. Furthermore, the system offers specialized productivity applications catering to specific industries or professions. For example, architecture, engineering, or graphic design professionals can utilize CAD (Computer-Aided Design) software, graphical editing tools, and 3D modeling applications, allowing them to work on designs, sketches, and renderings while in the vehicle. Similarly, entrepreneurs, freelancers, or small business owners can utilize accounting software, invoicing tools, or business management platforms to manage finances, track expenses, or monitor business performance from the convenience of their vehicle. The productivity tools are seamlessly integrated with the vehicle's navigation system, calendar, and personal assistant features. Occupants can access contextual information, set reminders, or schedule appointments without leaving the productivity environment. Moreover, the system leverages voice commands and gesture controls, allowing users to perform tasks hands-free.

[0070] In one embodiment, the system enhances the vehicle charging experience by intelligently selecting and transmitting content to the vehicle based on the remaining charging time. The system determines the remaining time until the vehicle's charging process is complete. This determination can be facilitated by monitoring the charging progress through the vehicle's charging system. Once the remaining time is established, the method selects and transmits relevant content to the vehicle based on this remaining time. The content selection process considers various factors, such as the preferences and interests of the vehicle occupants. This information is gathered through interactions with an application installed in the vehicle, which collects data on previous content usage, current interests, age, occupation, and other relevant demographics. Based on the gathered information, the system selects appropriate video, audio, or game content to share with the occupant during the charging session. Moreover, the method involves identifying actions performed by the occupant while using the application, particularly in interactive multimedia sessions like games. These actions include selections made during gameplay, such as jumping or moving quickly. In response to these actions, the method enables various vehicle functions to enhance the occupant's experience. For example, a rapid movement in the game might trigger an increase in air ventilation speed, ensuring the occupant feels a rush of air. Similarly, actions like punching or kicking in the game may prompt vibrations or seat movements to simulate the game's intensity within the vehicle. Furthermore, the method accounts for scenarios where a game session associated with the application ends prematurely due to the completion of the charging process. The system determines not to initiate another game session based on the updated remaining time. Instead, it may fill the remaining charging time with alternative content, such as streaming news headlines or other short-form content. Additionally, if the game session is not complete, the method ensures that the game session data is forwarded to the user's device for future access. This enables the occupant to resume the game on their device, leveraging a cloud server application where the

game profile is stored.

[0071] Flow diagrams depicted herein, such as FIG. 1A, FIG. 2C, FIG. 2D, FIG. 2E, and FIG. 2F, are separate examples but may be the same or different embodiments. Any of the operations in one flow diagram may be adopted and shared with another flow diagram. No example operation is intended to limit the subject matter of any embodiment or corresponding claim.

[0072] It is important to note that all the flow diagrams and corresponding processes derived from FIG. 1A, FIG. 2C, FIG. 2D, FIG. 2E, and FIG. 2F may be part of a same process or may share sub-processes with one another thus making the diagrams combinable into a single preferred embodiment that does not require any one specific operation but which performs certain operations from one example process and from one or more additional processes. All the example processes are related to the same physical system and can be used separately or interchangeably.

[0073] The instant solution can be used in conjunction with one or more types of vehicles: battery electric vehicles, hybrid vehicles, fuel cell vehicles, internal combustion engine vehicles and/or vehicles utilizing renewable sources.

[0074] FIG. 2A illustrates a vehicle network diagram **200**, according to example embodiments. The network comprises elements including a vehicle **202** including a processor **204**, as well as a vehicle **202'** including a processor **204'**. The vehicles **202**, **202'** communicate with one another via the processors **204**, **204'**, as well as other elements (not shown) including transceivers, transmitters, receivers, storage, sensors, and other elements capable of providing communication. The communication between the vehicles **202**, and **202'** can occur directly, via a private and/or a public network (not shown), or via other vehicles and elements comprising one or more of a processor, memory, and software. Although depicted as single vehicles and processors, a plurality of vehicles and processors may be present. One or more of the applications, features, steps, solutions, etc., described and/or depicted herein may be utilized and/or provided by the instant elements.

[0075] FIG. 2B illustrates another vehicle network diagram **210**, according to example embodiments. The network comprises elements including a vehicle **202** including a processor **204**, as well as a vehicle **202'** including a processor **204'**. The vehicles **202**, **202'** communicate with one another via the processors **204**, **204'**, as well as other elements (not shown), including transceivers, transmitters, receivers, storage, sensors, and other elements capable of providing communication. The communication between the vehicles **202**, and **202'** can occur directly, via a private and/or a public network (not shown), or via other vehicles and elements comprising one or more of a processor, memory, and software. The processors **204**, **204'** can further communicate with one or more elements **230** including sensor **212**, wired device **214**, wireless device **216**, database **218**, mobile phone **220**, vehicle **222**, computer **224**, input/output (I/O) device **226**, and voice application **228**. The processors **204**, **204'** can further communicate with elements comprising one or more of a processor, memory, and software.

[0076] Although depicted as single vehicles, processors and elements, a plurality of vehicles, processors and elements may be present. Information or communication can occur to and/or from any of the processors **204**, **204'** and elements **230**. For example, the mobile phone **220** may provide information to the processor **204**, which may initiate the vehicle **202** to take an action, may further provide the information or additional information to the processor **204'**, which may initiate the vehicle **202'** to take an action, may further provide the information or additional information to the mobile phone **220**, the vehicle **222**, and/or the computer **224**. One or more of the applications, features, steps, solutions, etc., described and/or depicted herein may be utilized and/or provided by the instant elements.

[0077] FIG. 2C illustrates yet another vehicle network diagram **240**, according to example embodiments. The network comprises elements including a vehicle **202**, a processor **204**, and a non-transitory computer readable medium **242C**. The processor **204** is communicably coupled to the non-transitory computer readable medium **242C** and elements **230** (which were depicted in FIG. 2B). The vehicle **202** may be a vehicle, server, or any device with a processor and memory.

[0078] The processor **204** performs one or more of determining a remaining time until charging of a vehicle is complete **244C** and transmitting content to the vehicle based on the remaining time **246C**.

[0079] FIG. 2D illustrates a further vehicle network diagram **250**, according to example embodiments. The network comprises elements including a vehicle **202**, a processor **204**, and a non-transitory computer readable medium **242D**. The processor **204** is communicably coupled to the non-transitory computer readable medium **242D** and elements **230** (which were depicted in FIG. 2B). The vehicle **202** may be a vehicle, server or any device with a processor and memory.

[0080] The processor **204** performs one or more of identifying an application associated with a profile of an occupant of the vehicle, and initiating the application while the vehicle is charging when the remaining time is within a time duration to use the application **244D**, identifying one or more actions performed by an occupant while using an application, and enabling one or more vehicle functions responsive to the one or more actions identified in the application **245D**, the one or more vehicle functions comprise a chair movement, a change in air ventilation speed or temperature, a change in light control, and a change in an air infuser **246D**, the content transmitted is streaming content of a game associated with an occupant of the vehicle **247D**, determining a game session associated with the application has ended prematurely at an updated remaining time, and determining not to initiate another game session based on the updated remaining time **248D**, determining the charging is complete, determining a game session associated with the application is not completed, and forwarding game session data to a user device **249D**.

[0081] While this example describes in detail only one vehicle **202**, multiple such nodes may be connected to the blockchain. It should be understood that the vehicle **202** may include additional components and that some of the components described herein may be removed and/or modified without departing from a scope of the instant application. The vehicle **202** may have a computing device or a server computer, or the like, and may include a processor **204**, which may be a semiconductor-based microprocessor, a central processing unit (CPU), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), and/or another hardware device. Although a single processor **204** is depicted, it should be understood that the vehicle **202** may include multiple processors, multiple cores, or the like without departing from the scope of the instant application. The vehicle **202** may be a vehicle, server or any device with a processor and memory.

[0082] The processor **204** performs one or more of receiving a confirmation of an event from one or more elements described or depicted herein, wherein the confirmation comprises a blockchain consensus between peers represented by any of the elements and executing a smart contract to record the confirmation on the blockchain consensus. Consensus is formed between one or more of any element **230** and/or any element described or depicted herein, including a vehicle, a server, a wireless device, etc. In another example, the vehicle **202** can be one or more of any element **230** and/or any element described or depicted herein, including a server, a wireless device, etc.

[0083] The processors and/or computer readable medium may fully or partially reside in the interior or exterior of the vehicles. The steps or features stored in the computer readable medium may be fully or partially performed by any of the processors and/or elements in any order. Additionally, one or more steps or features may be added, omitted, combined, performed at a later time, etc.

[0084] FIG. 2E illustrates a flow diagram **260**, according to example embodiments. Referring to FIG. 2E, the instant solution includes one or more of determining a remaining time until charging of a vehicle is complete **244E** and transmitting content to the vehicle based on the remaining time **246E**.

[0085] FIG. 2F illustrates another flow diagram **270**, according to example embodiments. Referring to FIG. 2F, the instant solution includes one or more of identifying an application associated with a profile of an occupant of the vehicle, and initiating the application while the vehicle is charging

when the remaining time is within a time duration to use the application **244F**, identifying one or more actions performed by an occupant while using an application, and enabling one or more vehicle functions responsive to the one or more actions identified in the application **245F**, the one or more vehicle functions comprise a chair movement, a change in air ventilation speed or temperature, a change in light control, and a change in an air infuser **246F**, the content transmitted is streaming content of a game associated with an occupant of the vehicle **247F**, determining a game session associated with the application has ended prematurely at an updated remaining time, and determining not to initiate another game session based on the updated remaining time **248F**, determining the charging is complete, determining a game session associated with the application is not completed, and forwarding game session data to a user device **249F**.

[0086] Technological advancements typically build upon the fundamentals of predecessor technologies; such is the case with Artificial Intelligence (AI) models. An AI classification system describes the stages of AI progression. The first classification is known as “Reactive Machines,” followed by present-day AI classification “Limited Memory Machines” (also known as “Artificial Narrow Intelligence”), then progressing to “Theory of Mind” (also known as “Artificial General Intelligence”), and reaching the AI classification “Self-Aware” (also known as “Artificial Superintelligence”). Present-day Limited Memory Machines are a growing group of AI models built upon the foundation of its predecessor, Reactive Machines. Reactive Machines emulate human responses to stimuli; however, they are limited in their capabilities as they cannot typically learn from prior experience. Once the AI model's learning abilities emerged, its classification was promoted to Limited Memory Machines. In this present-day classification, AI models learn from large volumes of data, detect patterns, solve problems, generate and predict data, and the like, while inheriting all of the capabilities of Reactive Machines. Examples of AI models classified as Limited Memory Machines include, but are not limited to, Chatbots, Virtual Assistants, Machine Learning (ML), Deep Learning (DL), Natural Language Processing (NLP), Generative AI (GenAI) models, and any future AI models that are yet to be developed possessing characteristics of Limited Memory Machines. Generative AI models combine Limited Memory Machine technologies, incorporating ML and DL, forming the foundational building blocks of future AI models. For example, Theory of Mind is the next progression of AI that may be able to perceive, connect, and react by generating appropriate reactions in response to an entity with which the AI model is interacting; all these capabilities rely on the fundamentals of Generative AI. Furthermore, in an evolution into the Self-Aware classification, AI models will be able to understand and evoke emotions in the entities they interact with, as well as possessing their own emotions, beliefs, and needs, all of which rely on Generative AI fundamentals of learning from experiences to generate and draw conclusions about itself and its surroundings. Generative AI models are integral and core to future artificial intelligence models. As described herein, Generative AI refers to present-day Generative AI models and future AI models.

[0087] FIG. 3A illustrates an AI/ML network diagram **300A** that supports AI-assisted vehicle or occupant decision points. Other branches of AI, such as, but not limited to, computer vision, fuzzy logic, expert systems, neural networks/deep learning, generative AI, and natural language processing, may all be employed in developing the AI model shown in these embodiments. Further, the AI model included in these embodiments is not limited to particular AI algorithms. Any algorithm or combination of algorithms related to supervised, unsupervised, and reinforcement learning algorithms may be employed.

[0088] In one embodiment, Generative AI (GenAI) may be used by the instant solution in the transformation of data. Vehicles are equipped with diverse sensors, cameras, radars, and LiDARs, which collect a vast array of data, such as images, speed readings, GPS data, and acceleration metrics. However, raw data, once acquired, undergoes preprocessing that may involve normalization, anonymization, missing value imputation, or noise reduction to allow the data to be further used effectively.

[0089] The GenAI executes data augmentation following the preprocessing of the data. Due to the limitation of datasets in capturing the vast complexity of real-world vehicle scenarios, augmentation tools are employed to expand the dataset. This might involve image-specific transformations like rotations, translations, or brightness adjustments. For non-image data, techniques like jittering can be used to introduce synthetic noise, simulating a broader set of conditions.

[0090] In the instant solution, data generation is then performed on the data. Tools like Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs) are trained on existing datasets to generate new, plausible data samples. For example, GANs might be tasked with crafting images showcasing vehicles in uncharted conditions or from unique perspectives. As another example, the synthesis of sensor data may be performed to model and create synthetic readings for such scenarios, enabling thorough system testing without actual physical encounters. A critical step in the use of GenAI, given the safety-critical nature of vehicles, is validation. This validation might include the output data being compared with real-world datasets or using specialized tools like a GAN discriminator to gauge the realism of the crafted samples.

[0091] Vehicle node **310** may include a plurality of sensors **312** that may include but are not limited to, light sensors, weight sensors, cameras, LiDAR, and radar. In some embodiments, these sensors **312** send data to a database **320** that stores data about the vehicle and occupants of the vehicle. In some embodiments, these sensors **312** send data to one or more decision subsystems **316** in vehicle node **310** to assist in decision-making.

[0092] Vehicle node **310** may include one or more user interfaces (UIs) **314**, such as a steering wheel, navigation controls, audio/video controls, temperature controls, etc. In some embodiments, these UIs **314** send data to a database **320** that stores event data about the UIs **314** that includes but is not limited to selection, state, and display data. In some embodiments, these UIs **314** send data to one or more decision subsystems **316** in vehicle node **310** to assist decision-making.

[0093] Vehicle node **310** may include one or more decision subsystems **316** that drive a decision-making process around, but are not limited to, vehicle control, temperature control, charging control, etc. In some embodiments, the decision subsystems **316** gather data from one or more sensors **312** to aid in the decision-making process. In some embodiments, a decision subsystem **316** may gather data from one or more UIs **314** to aid in the decision-making process. In some embodiments, a decision subsystem **316** may provide feedback to a UI **314**.

[0094] An AI/ML production system **330** may be used by a decision subsystem **316** in a vehicle node **310** to assist in its decision-making process. The AI/ML production system **330** includes one or more AI/ML models **332** that are executed to retrieve the needed data, such as, but not limited to, a prediction, a categorization, a UI prompt, etc. In some embodiments, an AI/ML production system **330** is hosted on a server. In some embodiments, the AI/ML production system **330** is cloud-hosted. In some embodiments, the AI/ML production system **330** is deployed in a distributed multi-node architecture. In some embodiments, the AI production system resides in vehicle node **310**.

[0095] An AI/ML development system **340** creates one or more AI/ML models **332**. In some embodiments, the AI/ML development system **340** utilizes data in the database **320** to develop and train one or more AI models **332**. In some embodiments, the AI/ML development system **340** utilizes feedback data from one or more AI/ML production systems **330** for new model development and/or existing model re-training. In an embodiment, the AI/ML development system **340** resides and executes on a server. In another embodiment the AI/ML development system **340** is cloud hosted. In a further embodiment, the AI/ML development system **340** utilizes a distributed data pipeline/analytics engine.

[0096] Once an AI/ML model **332** has been trained and validated in the AI/ML development system **340**, it may be stored in an AI/ML model registry **360** for retrieval by either the AI/ML development system **340** or by one or more AI/ML production systems **330**. The AI/ML model

registry **360** resides in a dedicated server in one embodiment. In some embodiments, the AI/ML model registry **360** is cloud-hosted. The AI/ML model registry **360** is a distributed database in other embodiments. In further embodiments, the AI/ML model registry **360** resides in the AI/ML production system **330**.

[0097] FIG. **3B** illustrates a process **300B** for developing one or more AI/ML models that support AI-assisted vehicle or occupant decision points. An AI/ML development system **340** executes steps to develop an AI/ML model **332** that begins with data extraction **342**, in which data is loaded and ingested from one or more data sources. In some embodiments, vehicle and user data is extracted from a database **320**. In some embodiments, model feedback data is extracted from one or more AI/ML production systems **330**.

[0098] Once the required data has been extracted **342**, it must be prepared **344** for model training. In some embodiments, this step involves statistical testing of the data to see how well it reflects real-world events, its distribution, the variety of data in the dataset, etc. In some embodiments, the results of this statistical testing may lead to one or more data transformations being employed to normalize one or more values in the dataset. In some embodiments, this step includes cleaning data deemed to be noisy. A noisy dataset includes values that do not contribute to the training, such as but are not limited to, null and long string values. Data preparation **344** may be a manual process or an automated process using one or more of the elements, functions described or depicted herein.

[0099] Features of the data are identified and extracted **346**. In some embodiments, a feature of the data is internal to the prepared data from step **344**. In other embodiments, a feature of the data requires a piece of prepared data from step **344** to be enriched by data from another data source to be useful in developing an AI/ML model **332**. In some embodiments, identifying features is a manual process or an automated process using one or more of the elements, functions described or depicted herein. Once the features have been identified, the values of the features are collected into a dataset that will be used to develop the AI/ML model **332**.

[0100] The dataset output from feature extraction step **346** is split **348** into a training and validation data set. The training data set is used to train the AI/ML model **332**, and the validation data set is used to evaluate the performance of the AI/ML model **332** on unseen data.

[0101] The AI/ML model **332** is trained and tuned **350** using the training data set from the data splitting step **348**. In this step, the training data set is fed into an AI/ML algorithm and an initial set of algorithm parameters. The performance of the AI/ML model **332** is then tested within the AI/ML development system **340** utilizing the validation data set from step **348**. These steps may be repeated with adjustments to one or more algorithm parameters until the model's performance is acceptable based on various goals and/or results.

[0102] The AI/ML model **332** is evaluated **352** in a staging environment (not shown) that resembles the ultimate AI/ML production system **330**. This evaluation uses a validation dataset to ensure the performance in an AI/ML production system **330** matches or exceeds expectations. In some embodiments, the validation dataset from step **348** is used. In other embodiments, one or more unseen validation datasets are used. In some embodiments, the staging environment is part of the AI/ML development system **340**. In other embodiments, the staging environment is managed separately from the AI/ML development system **340**. Once the AI/ML model **332** has been validated, it is stored in an AI/ML model registry **360**, which can be retrieved for deployment and future updates. As before, in some embodiments, the model evaluation step **352** is a manual process or an automated process using one or more of the elements, functions described or depicted herein.

[0103] Once an AI/ML model **332** has been validated and published to an AI/ML model registry **360**, it may be deployed **354** to one or more AI/ML production systems **330**. In some embodiments, the performance of deployed AI/ML models **332** is monitored **356** by the AI/ML development system **340**. In some embodiments, AI/ML model **332** feedback data is provided by the AI/ML production system **330** to enable model performance monitoring **356**. In some embodiments, the AI/ML development system **340** periodically requests feedback data for model performance

monitoring **356**. In some embodiments, model performance monitoring includes one or more triggers that result in the AI/ML model **332** being updated by repeating steps **342-354** with updated data from one or more data sources.

[0104] FIG. **3C** illustrates a process **300C** for utilizing an AI/ML model that supports AI-assisted vehicle or occupant decision points. As stated previously, the AI model utilization process depicted herein reflects ML, which is a particular branch of AI, but the instant solution is not limited to ML and is not limited to any AI algorithm or combination of algorithms.

[0105] Referring to FIG. **3C**, an AI/ML production system **330** may be used by a decision subsystem **316** in vehicle node **310** to assist in its decision-making process. The AI/ML production system **330** provides an application programming interface (API) **334**, executed by an AI/ML server process **336** through which requests can be made. In some embodiments, a request may include an AI/ML model **332** identifier to be executed. In some embodiments, the AI/ML model **332** to be executed is implicit based on the type of request. In some embodiments, a data payload (e.g., to be input to the model during execution) is included in the request. In some embodiments, the data payload includes sensor **312** data from vehicle node **310**. In some embodiments, the data payload includes UI **314** data from vehicle node **310**. In some embodiments, the data payload includes data from other vehicle node **310** subsystems (not shown), including but not limited to, occupant data subsystems. In an embodiment, one or more elements or nodes **320**, **330**, **340**, or **360** may be located in the vehicle node **310**.

[0106] Upon receiving the API **334** request, the AI/ML server process **336** may need to transform the data payload or portions of the data payload to be valid feature values into an AI/ML model **332**. Data transformation may include but is not limited to combining data values, normalizing data values, and enriching the incoming data with data from other data sources. Once any required data transformation occurs, the AI/ML server process **336** executes the appropriate AI/ML model **332** using the transformed input data. Upon receiving the execution result, the AI/ML server process **336** responds to the API caller, which is a decision subsystem **316** of vehicle node **310**. In some embodiments, the response may result in an update to a UI **314** in vehicle node **310**. In some embodiments, the response includes a request identifier that can be used later by the decision subsystem **316** to provide feedback on the AI/ML model **332** performance. Further, in some embodiments, immediate performance feedback may be recorded into a model feedback log **338** by the AI/ML server process **336**. In some embodiments, execution model failure is a reason for immediate feedback.

[0107] In some embodiments, the API **334** includes an interface to provide AI/ML model **332** feedback after an AI/ML model **332** execution response has been processed. This mechanism may be used to evaluate the performance of the AI/ML model **332** by enabling the API caller to provide feedback on the accuracy of the model results. For example, if the AI/ML model **332** provided an estimated time of arrival of 20 minutes, but the actual travel time was 24 minutes, that may be indicated. In some embodiments, the feedback interface includes the identifier of the initial request so that it can be used to associate the feedback with the request. Upon receiving a call into the feedback interface of API **334**, the AI/ML server process **336** records the feedback in the model feedback log **338**. In some embodiments, the data in this model feedback log **338** is provided to model performance monitoring **356** in the AI/ML development system **340**. This log data is streamed to the AI/ML development system **340** in one embodiment. In some embodiments, the log data is provided upon request.

[0108] A number of the steps/features that may utilize the AI/ML process described herein include one or more of: determining a remaining time until charging of a vehicle is complete, transmitting content to the vehicle based on the remaining time, identifying an application associated with a profile of an occupant of the vehicle, and initiating the application while the vehicle is charging when the remaining time is within a time duration to use the application, identifying one or more actions performed by an occupant while using an application, and enabling one or more vehicle

functions responsive to the one or more actions identified in the application, the one or more vehicle functions comprise a chair movement, a change in air ventilation speed or temperature, a change in light control, and a change in an air infuser, the content transmitted is streaming content of a game associated with an occupant of the vehicle, determining a game session associated with the application has ended prematurely at an updated remaining time, and determining not to initiate another game session based on the updated remaining time, determining the charging is complete, determining a game session associated with the application is not completed, and forwarding game session data to a user device.

[0109] Data associated with any of these steps/features, as well as any other features or functionality described or depicted herein, the AI/ML production system **330**, as well as one or more of the other elements depicted in FIG. **3C** may be used to process this data in a pre-transformation and/or post-transformation process. Data related to this process can be used by the vehicle node **310**. In one embodiment, data related to this process may be used with a charging station/charging point, a server, a wireless device, and/or any of the processors described or depicted herein.

[0110] FIG. **3D** illustrates a process **300D** of designing a new machine learning model via a user interface **370** of the system according to example embodiments. As an example, a model may be output as part of the AI/ML Development System **340**. Referring to FIG. **3D**, a user can use an input mechanism from a menu **372** of a user interface **370** to add pieces/components to a model being developed within a workspace **374** of the user interface **370**.

[0111] The menu **372** includes a plurality of graphical user interface (GUI) menu options which can be selected to reveal additional components that can be added to the model design shown in the workspace **374**. The GUI menu includes options for adding elements to the workspace, such as features which may include neural networks, machine learning models, AI models, data sources, conversion processes (e.g., vectorization, encoding, etc.), analytics, etc. The user can continue to add features to the model and connect them using edges or other elements to create a flow within the workspace **374**. For example, the user may add a node **376** to a flow of a new model within the workspace **374**. For example, the user may connect the node **376** to another node in the diagram via an edge **378**, creating a dependency within the diagram. When the user is done, the user can save the model for subsequent training/testing.

[0112] In another example, the name of the object can be identified from a web page or a user interface **370** where the object is visible within a browser or the workspace **374** on the user device. A pop-up within the browser or the workspace **374** can be overlaid where the object is visible, which includes an option to navigate to the identified web page corresponding to the alternative object via a rule set.

[0113] FIG. **3E** illustrates a process **300E** of accessing an object **392** from an object storage **390** of the host platform **380** according to example embodiments. For example, the object storage **390** may store data that is used by the AI models and machine learning (ML) models, training data, expected outputs for testing, training results, and the like. The object storage **390** may also store any other kind of data. Each object may include a unique identifier, a data section **394**, and a metadata section **396**, which provide a descriptive context associated with the data, including data that can later be extracted for purposes of machine learning. The unique identifier may uniquely identify an object with respect to all other objects in the object storage **390**. The data section **394** may include unstructured data such as web pages, digital content, images, audio, text, and the like.

[0114] Instead of breaking files into blocks stored on disks in a file system, the object storage **390** handles objects as discrete units of data stored in a structurally flat data environment. Here, the object storage may not use folders, directories, or complex hierarchies. Instead, each object may be a simple, self-contained repository that includes the data, the metadata, and the unique identifier that a client application can use to locate and access it. In this case, the metadata is more descriptive than a file-based approach. The metadata can be customized with additional context that

can later be extracted and leveraged for other purposes, such as data analytics.

[0115] The objects that are stored in the object storage **390** may be accessed via an API **384**. The API **384** may be a Hypertext Transfer Protocol (HTTP)-based RESTful API (also known as a RESTful Web service). The API **384** can be used by the client application to query an object's metadata to locate the desired object (data) via the Internet from anywhere on any device. The API **384** may use HTTP commands such as “PUT” or “POST” to upload an object, “GET” to retrieve an object, “DELETE” to remove an object, and the like.

[0116] The object storage **390** may provide a directory **398** that uses the metadata of the objects to locate appropriate data files. The directory **398** may contain descriptive information about each object stored in the object storage **390**, such as a name, a unique identifier, a creation timestamp, a collection name, etc. To query the object within the object storage **390**, the client application may submit a command, such as an HTTP command, with an identifier of the object **392**, a payload, etc. The object storage **390** can store the actions and results described herein, including associating two or more lists of ranked assets with one another based on variables used by the two or more lists of ranked assets that have a correlation above a predetermined threshold.

[0117] FIG. 4A illustrates a diagram **400A** depicting the electrification of one or more elements. In one example, a vehicle **402B** may provide power stored in its batteries to one or more elements, including other vehicle(s) **408B**, charging station(s) **406B**, and electric grid(s) **404B**. The electric grid(s) **404B** is/are coupled to one or more of the charging stations **406B**, which may be coupled to one or more of the vehicles **408B**. This configuration allows the distribution of electricity/power received from the vehicle **402B**. The vehicle **402B** may also interact with the other vehicle(s) **408B**, such as via V2V technology, communication over cellular, Wi-Fi, and the like. The vehicle **402B** may also interact wirelessly and/or wired with other vehicles **408B**, the charging station(s) **406B** and/or with the electric grid(s) **404B**. In one example, the vehicle **402B** is routed (or routes itself) in a safe and efficient manner to the electric grid(s) **404B**, the charging station(s) **406B**, or the other vehicle(s) **408B**. Using one or more embodiments of the instant solution, the vehicle **402B** can provide energy to one or more of the elements depicted herein in various advantageous ways as described and/or depicted herein. Further, the safety and efficiency of the vehicle may be increased, and the environment may be positively affected as described and/or depicted herein.

[0118] The term ‘energy’, ‘electricity’, ‘power’, and the like may be used to denote any form of energy received, stored, used, shared, and/or lost by the vehicle(s). The energy may be referred to in conjunction with a voltage source and/or a current supply of charge provided from an entity to the vehicle(s) during a charge/use operation. Energy may also be in the form of fossil fuels (for example, for use with a hybrid vehicle) or via alternative power sources, including but not limited to lithium-based, nickel-based, hydrogen fuel cells, atomic/nuclear energy, fusion-based energy sources, and energy generated during an energy sharing and/or usage operation for increasing or decreasing one or more vehicles energy levels at a given time.

[0119] In one example, the charging station **406B** manages the amount of energy transferred from the vehicle **402B** such that there is sufficient charge remaining in the vehicle **402B** to arrive at a destination. In one example, a wireless connection is used to wirelessly direct an amount of energy transfer between vehicles **408B**, wherein the vehicles may both be in motion. In one embodiment, wireless charging may occur via a fixed charger and batteries of the vehicle in alignment with one another (such as a charging mat in a garage or parking space). In one example, an idle vehicle, such as a vehicle **402B** (which may be autonomous) is directed to provide an amount of energy to a charging station **406B** and return to the original location (for example, its original location or a different destination). In one example, a mobile energy storage unit (not shown) is used to collect surplus energy from at least one other vehicle **408B** and transfer the stored surplus energy at a charging station **406B**. In one example, factors determine an amount of energy to transfer to a charging station **406B**, such as distance, time, as well as traffic conditions, road conditions, environmental/weather conditions, the vehicle's condition (weight, etc.), an occupant(s) schedule

while utilizing the vehicle, a prospective occupant(s) schedule waiting for the vehicle, etc. In one example, the vehicle(s) **408B**, the charging station(s) **406B** and/or the electric grid(s) **404B** can provide energy to the vehicle **402B**.

[0120] In one embodiment, a location such as a building, a residence, or the like (not depicted), communicably coupled to one or more of the electric grid **404B**, the vehicle **402B**, and/or the charging station(s) **406B**. The rate of electric flow to one or more of the location, the vehicle **402B**, the other vehicle(s) **408B** is modified, depending on external conditions, such as weather. For example, when the external temperature is extremely hot or extremely cold, raising the chance for an outage of electricity, the flow of electricity to a connected vehicle **402B/408B** is slowed to help minimize the chance for an outage.

[0121] In one embodiment, vehicles **402B** and **408B** may be utilized as bidirectional vehicles. Bidirectional vehicles are those that may serve as mobile microgrids that can assist in the supplying of electrical power to the grid **404B** and/or reduce the power consumption when the grid is stressed. Bidirectional vehicles incorporate bidirectional charging, which in addition to receiving a charge to the vehicle, the vehicle can transfer energy from the vehicle to the grid **404B**, otherwise referred to as “V2G”. In bidirectional charging, the electricity flows both ways; to the vehicle and from the vehicle. When a vehicle is charged, alternating current (AC) electricity from the grid **404B** is converted to direct current (DC). This may be performed by one or more of the vehicle's own converter or a converter on the charging station **406B**. The energy stored in the vehicle's batteries may be sent in an opposite direction back to the grid. The energy is converted from DC to AC through a converter usually located in the charging station **406B**, otherwise referred to as a bidirectional charger. Further, the instant solution as described and depicted with respect to FIG. **4B** can be utilized in this and other networks and/or systems.

[0122] FIG. **4B** is a diagram showing interconnections between different elements **400B**. The instant solution may be stored and/or executed entirely or partially on and/or by one or more computing devices **414C**, **418C**, **424C**, **428C**, **432C**, **436C**, **406C**, **442C** and **410C** associated with various entities, all communicably coupled and in communication with a network **402C**. A database **438C** is communicably coupled to the network and allows for the storage and retrieval of data. In one example, the database is an immutable ledger. One or more of the various entities may be a vehicle **404C**, one or more service provider **416C**, one or more public buildings **422C**, one or more traffic infrastructure **426C**, one or more residential dwellings **430C**, an electric grid/charging station **434C**, a microphone **440C**, and/or another vehicle **408C**. Other entities and/or devices, such as one or more private users using a smartphone **412C**, a laptop **420C**, an augmented reality (AR) device, a virtual reality (VR) device, and/or any wearable device may also interwork with the instant solution. The smartphone **412C**, laptop **420C**, the microphone **440C**, and other devices may be connected to one or more of the connected computing devices **414C**, **418C**, **424C**, **428C**, **432C**, **436C**, **406C**, **442C**, and **410C**. The one or more public buildings **422C** may include various agencies. The one or more public buildings **422C** may utilize a computing device **424C**. The one or more service provider **416C** may include a dealership, a tow truck service, a collision center, or other repair shop. The one or more service provider **416C** may utilize a computing apparatus **418C**. These various computer devices may be directly and/or communicably coupled to one another, such as via wired networks, wireless networks, blockchain networks, and the like. The microphone **440C** may be utilized as a virtual assistant, in one example. In one example, the one or more traffic infrastructure **426C** may include one or more traffic signals, one or more sensors including one or more cameras, vehicle speed sensors or traffic sensors, and/or other traffic infrastructure. The one or more traffic infrastructure **426C** may utilize a computing device **428C**.

[0123] In one embodiment, anytime an electrical charge is given or received to/from a charging station and/or an electrical grid, the entities that allow that to occur are one or more of a vehicle, a charging station, a server, and a network communicably coupled to the vehicle, the charging station, and the electrical grid.

[0124] In one example, a vehicle **408C/404C** can transport a person, an object, a permanently or temporarily affixed apparatus, and the like. In one example, the vehicle **408C** may communicate with vehicle **404C** via V2V communication through the computers associated with each vehicle **406C** and **410C** and may be referred to as a car, vehicle, automobile, and the like. The vehicle **404C/408C** may be a self-propelled wheeled conveyance, such as a car, a sports utility vehicle, a truck, a bus, a van, or other motor or battery-driven or fuel cell-driven vehicle. For example, vehicle **404C/408C** may be an electric vehicle, a hybrid vehicle, a hydrogen fuel cell vehicle, a plug-in hybrid vehicle, or any other type of vehicle with a fuel cell stack, a motor, and/or a generator. Other examples of vehicles include bicycles, scooters, trains, planes, boats, and any other form of conveyance that is capable of transportation. The vehicle **404C/408C** may be semi-autonomous or autonomous. For example, vehicle **404C/408C** may be self-maneuvering and navigate without human input. An autonomous vehicle may have and use one or more sensors and/or a navigation unit to drive autonomously. All of the data described or depicted herein can be stored, analyzed, processed and/or forwarded by one or more of the elements in FIG. 4B.

[0125] FIG. 4C is another block diagram showing interconnections between different elements in one example **400C**. A vehicle **412D** is presented and includes ECUs **410D**, **408D**, and a Head Unit (otherwise known as an Infotainment System) **406D**. An ECU is an embedded system in automotive electronics controlling one or more of the electrical systems or subsystems in a vehicle. ECUs may include but are not limited to the management of a vehicle's engine, brake system, gearbox system, door locks, dashboard, airbag system, infotainment system, electronic differential, and active suspension. ECUs are connected to the vehicle's Controller Area Network (CAN) bus **416D**. The ECUs may also communicate with a vehicle computer **404D** via the CAN bus **416D**. The vehicle's processors/sensors (such as the vehicle computer) **404D** can communicate with external elements, such as a server **418D** via a network **402D** (such as the Internet). Each ECU **410D**, **408D**, and Head Unit **406D** may contain its own security policy. The security policy defines permissible processes that can be executed in the proper context. In one example, the security policy may be partially or entirely provided in the vehicle computer **404D**.

[0126] ECUs **410D**, **408D**, and Head Unit **406D** may each include a custom security functionality element **414D** defining authorized processes and contexts within which those processes are permitted to run. Context-based authorization to determine validity if a process can be executed allows ECUs to maintain secure operation and prevent unauthorized access from elements such as the vehicle's CAN Bus. When an ECU encounters a process that is unauthorized, that ECU can block the process from operating. Automotive ECUs can use different contexts to determine whether a process is operating within its permitted bounds, such as proximity contexts, nearby objects, distance to approaching objects, speed, and trajectory relative to other moving objects, and operational contexts such as an indication of whether the vehicle is moving or parked, the vehicle's current speed, the transmission state, user-related contexts such as devices connected to the transport via wireless protocols, use of the infotainment, cruise control, parking assist, driving assist, location-based contexts, and/or other contexts.

[0127] Referring to FIG. 4D, an operating environment **400D** for a connected vehicle, is illustrated according to some embodiments. As depicted, the vehicle **410E** includes a CAN bus **408E** connecting elements **412E-426E** of the vehicle. Other elements may be connected to the CAN bus and are not depicted herein. The depicted elements connected to the CAN bus include a sensor set **412E**, Electronic Control Units **414E**, autonomous features or Advanced Driver Assistance Systems (ADAS) **416E**, and the navigation system **418E**. In some embodiments, the vehicle **410E** includes a processor **420E**, a memory **422E**, a communication unit **424E**, and an electronic display **426E**.

[0128] The processor **420E** includes an arithmetic logic unit, a microprocessor, a general-purpose controller, and/or a similar processor array to perform computations and provide electronic display signals to a display unit **426E**. The processor **420E** processes data signals and may include various computing architectures, including a complex instruction set computer (CISC) architecture, a

reduced instruction set computer (RISC) architecture, or an architecture implementing a combination of instruction sets. The vehicle **410E** may include one or more processors **420E**. Other processors, operating systems, sensors, displays, and physical configurations that are communicably coupled to one another (not depicted) may be used with the instant solution.

[0129] Memory **422E** is a non-transitory memory storing instructions or data that may be accessed and executed by the processor **420E**. The instructions and/or data may include code to perform the techniques described herein. The memory **422E** may be a dynamic random-access memory (DRAM) device, a static random-access memory (SRAM) device, flash memory, or another memory device. In some embodiments, the memory **422E** also may include non-volatile memory or a similar permanent storage device and media, which may include a hard disk drive, a floppy disk drive, a compact disc read only memory (CD-ROM) device, a digital versatile disk read only memory (DVD-ROM) device, a digital versatile disk random access memory (DVD-RAM) device, a digital versatile disk rewritable (DVD-RW) device, a flash memory device, or some other mass storage device for storing information on a permanent basis. A portion of the memory **422E** may be reserved for use as a buffer or virtual random-access memory (virtual RAM). The vehicle **410E** may include one or more memories **422E** without deviating from the current solution.

[0130] The memory **422E** of the vehicle **410E** may store one or more of the following types of data: navigation route data **418E**, and autonomous features data **416E**. In some embodiments, the memory **422E** stores data that may be necessary for the navigation application **418E** to provide the functions.

[0131] The navigation system **418E** may describe at least one navigation route including a start point and an endpoint. In some embodiments, the navigation system **418E** of the vehicle **410E** receives a request from a user for navigation routes wherein the request includes a starting point and an ending point. The navigation system **418E** may query a real-time data server **404E** (via a network **402E**), such as a server that provides driving directions, for navigation route data corresponding to navigation routes, including the start point and the endpoint. The real-time data server **404E** transmits the navigation route data to the vehicle **410E** via a wireless network **402E**, and the communication system **424E** stores the navigation data **418E** in the memory **422E** of the vehicle **410E**.

[0132] The ECU **414E** controls the operation of many of the systems of the vehicle **410E**, including the ADAS systems **416E**. The ECU **414E** may, responsive to instructions received from the navigation system **418E**, deactivate any unsafe and/or unselected autonomous features for the duration of a journey controlled by the ADAS systems **416E**. In this way, the navigation system **418E** may control whether ADAS systems **416E** are activated or enabled so that they may be activated for a given navigation route.

[0133] The sensor set **412E** may include any sensors in the vehicle **410E** generating sensor data. For example, the sensor set **412E** may include short-range sensors and long-range sensors. In some embodiments, the sensor set **412E** of the vehicle **410E** may include one or more of the following vehicle sensors: a camera, a Light Detection and Ranging (LiDAR) sensor, an ultrasonic sensor, an automobile engine sensor, a radar sensor, a laser altimeter, a manifold absolute pressure sensor, an infrared detector, a motion detector, a thermostat, a sound detector, a carbon monoxide sensor, a carbon dioxide sensor, an oxygen sensor, a mass airflow sensor, an engine coolant temperature sensor, a throttle position sensor, a crankshaft position sensor, a valve timer, an air-fuel ratio meter, a blind spot meter, a curb feeler, a defect detector, a Hall effect sensor, a parking sensor, a radar gun, a speedometer, a speed sensor, a tire-pressure monitoring sensor, a torque sensor, a transmission fluid temperature sensor, a turbine speed sensor (TSS), a variable reluctance sensor, a vehicle speed sensor (VSS), a water sensor, a wheel speed sensor, a global positioning system (GPS) sensor, a mapping functionality, and any other type of automotive sensor. The navigation system **418E** may store the sensor data in the memory **422E**.

[0134] The communication unit **424E** transmits and receives data to and from the network **402E** or

to another communication channel. In some embodiments, the communication unit **424E** may include a dedicated short-range communication (DSRC) transceiver, a DSRC receiver, and other hardware or software necessary to make the vehicle **410E** a DSRC-equipped device.

[0135] The vehicle **410E** may interact with other vehicles **406E** via V2V technology. V2V communication includes sensing radar information corresponding to relative distances to external objects, receiving GPS information of the vehicles, setting areas where the other vehicles **406E** are located based on the sensed radar information, calculating probabilities that the GPS information of the object vehicles will be located at the set areas, and identifying vehicles and/or objects corresponding to the radar information and the GPS information of the object vehicles based on the calculated probabilities, in one example.

[0136] For a vehicle to be adequately secured, the vehicle must be protected from unauthorized physical access as well as unauthorized remote access (e.g., cyber-threats). To prevent unauthorized physical access, a vehicle is equipped with a secure access system such as a keyless entry in one example. Meanwhile, security protocols are added to a vehicle's computers and computer networks to facilitate secure remote communications to and from the vehicle in one example.

[0137] ECUs are nodes within a vehicle that control tasks such as activating the windshield wipers to tasks such as an anti-lock brake system. ECUs are often connected to one another through the vehicle's central network, which may be referred to as a controller area network (CAN). State-of-the-art features such as autonomous driving are strongly reliant on implementing new, complex ECUs such as ADAS, sensors, and the like. While these new technologies have helped improve the safety and driving experience of a vehicle, they have also increased the number of externally-communicating units inside of the vehicle, making them more vulnerable to attack. Below are some examples of protecting the vehicle from physical intrusion and remote intrusion.

[0138] In one embodiment, a CAN includes a CAN bus with a high and low terminal and a plurality of ECUs, which are connected to the CAN bus via wired connections. The CAN bus is designed to allow microcontrollers and devices to communicate with each other in an application without a host computer. The CAN bus implements a message-based protocol (i.e., ISO 11898 standards) that allows ECUs to send commands to one another at a root level. Meanwhile, the ECUs represent controllers for controlling electrical systems or subsystems within the vehicle. Examples of the electrical systems include power steering, anti-lock brakes, air-conditioning, tire pressure monitoring, cruise control, and many other features.

[0139] In this example, the ECU includes a transceiver and a microcontroller. The transceiver may be used to transmit and receive messages to and from the CAN bus. For example, the transceiver may convert the data from the microcontroller into a format of the CAN bus and also convert data from the CAN bus into a format for the microcontroller. Meanwhile, the microcontroller interprets the messages and also decides what messages to send using ECU software installed therein in one example.

[0140] To protect the CAN from cyber threats, various security protocols may be implemented. For example, sub-networks (e.g., sub-networks A and B, etc.) may be used to divide the CAN into smaller sub-CANs and limit an attacker's capabilities to access the vehicle remotely. In one embodiment, a firewall (or gateway, etc.) may be added to block messages from crossing the CAN bus across sub-networks. If an attacker gains access to one sub-network, the attacker will not have access to the entire network. To make sub-networks even more secure, the most critical ECUs are not placed on the same sub-network, in one example.

[0141] In addition to protecting a vehicle's internal network, vehicles may also be protected when communicating with external networks such as the Internet. One of the benefits of having a vehicle connection to a data source such as the Internet is that information from the vehicle can be sent through a network to remote locations for analysis. Examples of vehicle information include GPS, onboard diagnostics, tire pressure, and the like. These communication systems are often referred to as telematics because they involve the combination of telecommunications and informatics.

Further, the instant solution as described and depicted can be utilized in this and other networks and/or systems, including those that are described and depicted herein.

[0142] FIG. 4E illustrates an example 400E of vehicles 402I and 408I performing secured V2V communications using security certificates, according to example embodiments. Referring to FIG. 4E, the vehicles 402I and 408I may communicate via V2V communications over a short-range network, a cellular network, or the like. Before sending messages, the vehicles 402I and 408I may sign the messages using a respective public key certificate. For example, the vehicle 402I may sign a V2V message using a public key certificate 404I. Likewise, the vehicle 408I may sign a V2V message using a public key certificate 410I. The public key certificates 404I and 410I are associated with the vehicles 402I and 408I, respectively, in one example.

[0143] Upon receiving the communications from each other, the vehicles may verify the signatures with a certificate authority 406I or the like. For example, the vehicle 408I may verify with the certificate authority 406I that the public key certificate 404I used by vehicle 402I to sign a V2V communication is authentic. If the vehicle 408I successfully verifies the public key certificate 404I, the vehicle knows that the data is from a legitimate source. Likewise, the vehicle 402I may verify with the certificate authority 406I that the public key certificate 410I used by the vehicle 408I to sign a V2V communication is authentic. Further, the instant solution as described and depicted with respect to FIG. 4E can be utilized in this and other networks and/or systems including those that are described and depicted herein.

[0144] In some embodiments, a computer may include a security processor. In particular, the security processor may perform authorization, authentication, cryptography (e.g., encryption), and the like, for data transmissions that are sent between ECUs and other devices on a CAN bus of a vehicle, and also data messages that are transmitted between different vehicles. The security processor may include an authorization module, an authentication module, and a cryptography module. The security processor may be implemented within the vehicle's computer and may communicate with other vehicle elements, for example, the ECUs/CAN network, wired and wireless devices such as wireless network interfaces, input ports, and the like. The security processor may ensure that data frames (e.g., CAN frames, etc.) that are transmitted internally within a vehicle (e.g., via the ECUs/CAN network) are secure. Likewise, the security processor can ensure that messages transmitted between different vehicles and devices attached or connected via a wire to the vehicle's computer are also secured.

[0145] For example, the authorization module may store passwords, usernames, PIN codes, biometric scans, and the like for different vehicle users. The authorization module may determine whether a user (or technician) has permission to access certain settings such as a vehicle's computer. In some embodiments, the authorization module may communicate with a network interface to download any necessary authorization information from an external server. When a user desires to make changes to the vehicle settings or modify technical details of the vehicle via a console or GUI within the vehicle or via an attached/connected device, the authorization module may require the user to verify themselves in some way before such settings are changed. For example, the authorization module may require a username, a password, a PIN code, a biometric scan, a predefined line drawing or gesture, and the like. In response, the authorization module may determine whether the user has the necessary permissions (access, etc.) being requested.

[0146] The authentication module may be used to authenticate internal communications between ECUs on the CAN network of the vehicle. As an example, the authentication module may provide information for authenticating communications between the ECUs. As an example, the authentication module may transmit a bit signature algorithm to the ECUs of the CAN network. The ECUs may use the bit signature algorithm to insert authentication bits into the CAN fields of the CAN frame. All ECUs on the CAN network typically receive each CAN frame. The bit signature algorithm may dynamically change the position, amount, etc., of authentication bits each time a new CAN frame is generated by one of the ECUs. The authentication module may also

provide a list of ECUs that are exempt (safe list) and that do not need to use the authentication bits. The authentication module may communicate with a remote server to retrieve updates to the bit signature algorithm and the like.

[0147] The encryption module may store asymmetric key pairs to be used by the vehicle to communicate with other external user devices and vehicles. For example, the encryption module may provide a private key to be used by the vehicle to encrypt/decrypt communications, while the corresponding public key may be provided to other user devices and vehicles to enable the other devices to decrypt/encrypt the communications. The encryption module may communicate with a remote server to receive new keys, updates to keys, keys of new vehicles, users, etc., and the like. The encryption module may also transmit any updates to a local private/public key pair to the remote server.

[0148] FIG. 5A illustrates an example vehicle configuration **500A** for managing database transactions associated with a vehicle, according to example embodiments. Referring to FIG. 5A, as a particular vehicle **525A** is engaged in transactions (e.g., vehicle service, dealer transactions, delivery/pickup, transportation services, etc.), the vehicle may receive assets **510A** and/or expel/transfer assets **512A** according to a transaction(s). A vehicle processor **526A** resides in the vehicle **525A** and communication exists between the vehicle processor **526A**, a database **530A**, and the transaction module **520A**. The transaction module **520A** may record information, such as assets, parties, credits, service descriptions, date, time, location, results, notifications, unexpected events, etc. Those transactions in the transaction module **520A** may be replicated into a database **530A**. The database **530A** can be one of a SQL database, a relational database management system (RDBMS), a relational database, a non-relational database, a blockchain, a distributed ledger, and may be on board the vehicle, may be off-board the vehicle, may be accessed directly and/or through a network, or be accessible to the vehicle.

[0149] In one embodiment, a vehicle may engage with another vehicle to perform various actions such as to share, transfer, acquire service calls, etc. when the vehicle has reached a status where the services need to be shared with another vehicle. For example, the vehicle may be due for a battery charge and/or may have an issue with a tire and may be in route to pick up a package for delivery. A vehicle processor resides in the vehicle and communication exists between the vehicle processor, a first database, and a transaction module. The vehicle may notify another vehicle, which is in its network and which operates on its blockchain member service. A vehicle processor resides in another vehicle and communication exists between the vehicle processor, a second database, the vehicle processor, and a transaction module. The another vehicle may then receive the information via a wireless communication request to perform the package pickup from the vehicle and/or from a server (not shown). The transactions are logged in the transaction modules and of both vehicles. The credits are transferred from the vehicle to the other vehicle and the record of the transferred service is logged in the first database, assuming that the blockchains are different from one another, or are logged in the same blockchain used by all members. The first database can be one of a SQL database, an RDBMS, a relational database, a non-relational database, a blockchain, a distributed ledger, and may be on board the vehicle, may be off-board the vehicle, may be accessible directly and/or through a network.

[0150] FIG. 5B illustrates a blockchain architecture configuration **500B**, according to example embodiments. Referring to FIG. 5B, the blockchain architecture **500B** may include certain blockchain elements, for example, a group of blockchain member nodes **502B-505B** as part of a blockchain group **510B**. In one example embodiment, a permissioned blockchain is not accessible to all parties but only to those members with permissioned access to the blockchain data. The blockchain nodes participate in a number of activities, such as blockchain entry addition and validation process (consensus). One or more of the blockchain nodes may endorse entries based on an endorsement policy and may provide an ordering service for all blockchain nodes. A blockchain node may initiate a blockchain action (such as an authentication) and seek to write to a blockchain

immutable ledger stored in the blockchain, a copy of which may also be stored on the underpinning physical infrastructure.

[0151] The blockchain transactions **520B** are stored in memory of computers as the transactions are received and approved by the consensus model dictated by the members' nodes. Approved transactions **526B** are stored in current blocks of the blockchain and committed to the blockchain via a committal procedure, which includes performing a hash of the data contents of the transactions in a current block and referencing a previous hash of a previous block. Within the blockchain, one or more smart contracts **530B** may exist that define the terms of transaction agreements and actions included in smart contract executable application code **532B**, such as registered recipients, vehicle features, requirements, permissions, sensor thresholds, etc. The code may be configured to identify whether requesting entities are registered to receive vehicle services, what service features they are entitled/required to receive given their profile statuses and whether to monitor their actions in subsequent events. For example, when a service event occurs and a user is riding in the vehicle, the sensor data monitoring may be triggered, and a certain parameter, such as a vehicle charge level, may be identified as being above/below a particular threshold for a particular period of time, then the result may be a change to a current status, which requires an alert to be sent to the managing party (i.e., vehicle owner, vehicle operator, server, etc.) so the service can be identified and stored for reference. The vehicle sensor data collected may be based on types of sensor data used to collect information about vehicle's status. The sensor data may also be the basis for the vehicle event data **534B**, such as a location(s) to be traveled, an average speed, a top speed, acceleration rates, whether there were any collisions, was the expected route taken, what is the next destination, whether safety measures are in place, whether the vehicle has enough charge/fuel, etc. All such information may be the basis of smart contract terms **530B**, which are then stored in a blockchain. For example, sensor thresholds stored in the smart contract can be used as the basis for whether a detected service is necessary and when and where the service should be performed.

[0152] In one embodiment, a blockchain logic example includes a blockchain application interface as an application programming interface (API) or plug-in application that links to the computing device and execution platform for a particular transaction. The blockchain configuration may include one or more applications, which are linked to APIs to access and execute stored program/application code (e.g., smart contract executable code, smart contracts, etc.), which can be created according to a customized configuration sought by participants and can maintain their own state, control their own assets, and receive external information. This can be deployed as an entry and installed, via appending to the distributed ledger, on all blockchain nodes.

[0153] The smart contract application code provides a basis for the blockchain transactions by establishing application code, which when executed causes the transaction terms and conditions to become active. The smart contract, when executed, causes certain approved transactions to be generated, which are then forwarded to the blockchain platform. The platform includes a security/authorization, computing devices, which execute the transaction management and a storage portion as a memory that stores transactions and smart contracts in the blockchain.

[0154] The blockchain platform may include various layers of blockchain data, services (e.g., cryptographic trust services, virtual execution environment, etc.), and underpinning physical computer infrastructure that may be used to receive and store new entries and provide access to auditors, which are seeking to access data entries. The blockchain may expose an interface that provides access to the virtual execution environment necessary to process the program code and engage the physical infrastructure. Cryptographic trust services may be used to verify entries such as asset exchange entries and keep information private.

[0155] The blockchain architecture configuration of FIGS. 5A and 5B may process and execute program/application code via one or more interfaces exposed, and services provided, by the blockchain platform. As a non-limiting example, smart contracts may be created to execute

reminders, updates, and/or other notifications subject to the changes, updates, etc. The smart contracts can themselves be used to identify rules associated with authorization and access requirements and usage of the ledger. For example, the information may include a new entry, which may be processed by one or more processing entities (e.g., processors, virtual machines, etc.) included in the blockchain layer. The result may include a decision to reject or approve the new entry based on the criteria defined in the smart contract and/or a consensus of the peers. The physical infrastructure may be utilized to retrieve any of the data or information described herein.

[0156] Within smart contract executable code, a smart contract may be created via a high-level application and programming language, and then written to a block in the blockchain. The smart contract may include executable code that is registered, stored, and/or replicated with a blockchain (e.g., distributed network of blockchain peers). An entry is an execution of the smart contract code, which can be performed in response to conditions associated with the smart contract being satisfied. The executing of the smart contract may trigger a trusted modification(s) to a state of a digital blockchain ledger. The modification(s) to the blockchain ledger caused by the smart contract execution may be automatically replicated throughout the distributed network of blockchain peers through one or more consensus protocols.

[0157] The smart contract may write data to the blockchain in the format of key-value pairs. Furthermore, the smart contract code can read the values stored in a blockchain and use them in application operations. The smart contract code can write the output of various logic operations into the blockchain. The code may be used to create a temporary data structure in a virtual machine or other computing platform. Data written to the blockchain can be public and/or can be encrypted and maintained as private. The temporary data that is used/generated by the smart contract is held in memory by the supplied execution environment, then deleted once the data needed for the blockchain is identified.

[0158] A smart contract executable code may include the code interpretation of a smart contract, with additional features. As described herein, the smart contract executable code may be program code deployed on a computing network, where it is executed and validated by chain validators together during a consensus process. The smart contract executable code receives a hash and retrieves from the blockchain a hash associated with the data template created by use of a previously stored feature extractor. If the hashes of the hash identifier and the hash created from the stored identifier template data match, then the smart contract executable code sends an authorization key to the requested service. The smart contract executable code may write to the blockchain data associated with the cryptographic details.

[0159] FIG. 5C illustrates a blockchain configuration for storing blockchain transaction data, according to example embodiments. Referring to FIG. 5C, the example configuration **500C** provides for the vehicle **562C**, the user device **564C** and a server **566C** sharing information with a distributed ledger (i.e., blockchain) **568C**. The server may represent a service provider entity inquiring with a vehicle service provider to share user profile rating information in the event that a known and established user profile is attempting to rent a vehicle with an established rated profile. The server **566C** may be receiving and processing data related to a vehicle's service requirements. As the service events occur, such as the vehicle sensor data indicates a need for fuel/charge, a maintenance service, etc., a smart contract may be used to invoke rules, thresholds, sensor information gathering, etc., which may be used to invoke the vehicle service event. The blockchain transaction data **570C** is saved for each transaction, such as the access event, the subsequent updates to a vehicle's service status, event updates, etc. The transactions may include the parties, the requirements (e.g., 18 years of age, service eligible candidate, valid driver's license, etc.), compensation levels, the distance traveled during the event, the registered recipients permitted to access the event and host a vehicle service, rights/permissions, sensor data retrieved during the vehicle event operation to log details of the next service event and identify a vehicle's condition status, and thresholds used to make determinations about whether the service event was completed

and whether the vehicle's condition status has changed.

[0160] FIG. 5D illustrates blockchain blocks that can be added to a distributed ledger, according to example embodiments, and contents of block structures 582A to 582n. Referring to FIG. 5D, clients (not shown) may submit entries to blockchain nodes to enact activity on the blockchain. As an example, clients may be applications that act on behalf of a requester, such as a device, person, or entity to propose entries for the blockchain. The plurality of blockchain peers (e.g., blockchain nodes) may maintain a state of the blockchain network and a copy of the distributed ledger.

Different types of blockchain nodes/peers may be present in the blockchain network including endorsing peers, which simulate and endorse entries proposed by clients and committing peers which verify endorsements, validate entries, and commit entries to the distributed ledger. In this example, the blockchain nodes may perform the role of endorser node, committer node, or both.

[0161] The instant system includes a blockchain that stores immutable, sequenced records in blocks, and a state database (current world state) maintaining a current state of the blockchain. One distributed ledger may exist per channel and each peer maintains its own copy of the distributed ledger for each channel of which they are a member. The instant blockchain is an entry log, structured as hash-linked blocks where each block contains a sequence of N entries. Blocks may include various components such as those shown in FIG. 5D. The linking of the blocks may be generated by adding a hash of a prior block's header within a block header of a current block. In this way, all entries on the blockchain are sequenced and cryptographically linked together preventing tampering with blockchain data without breaking the hash links. Furthermore, because of the links, the latest block in the blockchain represents every entry that has come before it. The instant blockchain may be stored on a peer file system (local or attached storage), which supports an append-only blockchain workload.

[0162] The current state of the blockchain and the distributed ledger may be stored in the state database. Here, the current state data represents the latest values for all keys ever included in the chain entry log of the blockchain. Smart contract executable code invocations execute entries against the current state in the state database. To make these smart contract executable code interactions extremely efficient, the latest values of all keys are stored in the state database. The state database may include an indexed view into the entry log of the blockchain, it can therefore be regenerated from the chain at any time. The state database may automatically get recovered (or generated if needed) upon peer startup before entries are accepted.

[0163] Endorsing nodes receive entries from clients and endorse the entry based on simulated results. Endorsing nodes hold smart contracts, which simulate the entry proposals. When an endorsing node endorses an entry, the endorsing nodes creates an entry endorsement, which is a signed response from the endorsing node to the client application indicating the endorsement of the simulated entry. The method of endorsing an entry depends on an endorsement policy that may be specified within smart contract executable code. An example of an endorsement policy is “the majority of endorsing peers must endorse the entry.” Different channels may have different endorsement policies. Endorsed entries are forwarded by the client application to an ordering service.

[0164] The ordering service accepts endorsed entries, orders them into a block, and delivers the blocks to the committing peers. For example, the ordering service may initiate a new block when a threshold of entries has been reached, a timer times out, or another condition. In this example, blockchain node is a committing peer that has received a data block 582A for storage on the blockchain. The ordering service may be made up of a cluster of orderers. The ordering service does not process entries, smart contracts, or maintain the shared ledger. Rather, the ordering service may accept the endorsed entries and specifies the order in which those entries are committed to the distributed ledger. The architecture of the blockchain network may be designed such that the specific implementation of ‘ordering’ becomes a pluggable component.

[0165] Entries are written to the distributed ledger in a consistent order. The order of entries is

established to ensure that the updates to the state database are valid when they are committed to the network. Unlike a cryptocurrency blockchain system where ordering occurs through the solving of a cryptographic puzzle, or mining, in this example the parties of the distributed ledger may choose the ordering mechanism that best suits that network.

[0166] Referring to FIG. 5D, a block **582A** (also referred to as a data block) that is stored on the blockchain and/or the distributed ledger may include multiple data segments such as a block header **584A** to **584n**, transaction-specific data **586A** to **586n**, and block metadata **588A** to **588n**. It should be appreciated that the various depicted blocks and their contents, such as block **582A** and its contents are merely for purposes of an example and are not meant to limit the scope of the example embodiments. In some cases, both the block header **584A** and the block metadata **588A** may be smaller than the transaction-specific data **586A**, which stores entry data; however, this is not a requirement. The block **582A** may store transactional information of N entries (e.g., **100**, **500**, **1000**, **2000**, **3000**, etc.) within the block data **590A** to **590n**. The block **582A** may also include a link to a previous block (e.g., on the blockchain) within the block header **584A**. In particular, the block header **584A** may include a hash of a previous block's header. The block header **584A** may also include a unique block number, a hash of the block data **590A** of the current block **582A**, and the like. The block number of the block **582A** may be unique and assigned in an incremental/sequential order starting from zero. The first block in the blockchain may be referred to as a genesis block, which includes information about the blockchain, its members, the data stored therein, etc.

[0167] The block data **590A** may store entry information of each entry that is recorded within the block. For example, the entry data may include one or more of a type of the entry, a version, a timestamp, a channel ID of the distributed ledger, an entry ID, an epoch, a payload visibility, a smart contract executable code path (deploy tx), a smart contract executable code name, a smart contract executable code version, input (smart contract executable code and functions), a client (creator) identify such as a public key and certificate, a signature of the client, identities of endorsers, endorser signatures, a proposal hash, smart contract executable code events, response status, namespace, a read set (list of key and version read by the entry, etc.), a write set (list of key and value, etc.), a start key, an end key, a list of keys, a Merkel tree query summary, and the like. The entry data may be stored for each of the N entries.

[0168] In some embodiments, the block data **590A** may also store transaction-specific data **586A**, which adds additional information to the hash-linked chain of blocks in the blockchain. Accordingly, the data **586A** can be stored in an immutable log of blocks on the distributed ledger. Some of the benefits of storing such data **586A** are reflected in the various embodiments disclosed and depicted herein. The block metadata **588A** may store multiple fields of metadata (e.g., as a byte array, etc.). Metadata fields may include signature on block creation, a reference to a last configuration block, an entry filter identifying valid and invalid entries within the block, last offset persisted of an ordering service that ordered the block, and the like. The signature, the last configuration block, and the orderer metadata may be added by the ordering service. Meanwhile, a committer of the block (such as a blockchain node) may add validity/invalidity information based on an endorsement policy, verification of read/write sets, and the like. The entry filter may include a byte array of a size equal to the number of entries in the block data and a validation code identifying whether an entry was valid/invalid.

[0169] The other blocks **582B** to **582n** in the blockchain also have headers, files, and values. However, unlike the first block **582A**, each of the headers **584A** to **584n** in the other blocks includes the hash value of an immediately preceding block. The hash value of the immediately preceding block may be just the hash of the header of the previous block or may be the hash value of the entire previous block. By including the hash value of a preceding block in each of the remaining blocks, a trace can be performed from the Nth block back to the genesis block (and the associated original file) on a block-by-block basis, as indicated by arrows **592**, to establish an

auditable and immutable chain-of-custody.

[0170] FIG. 5E illustrates a process 500E of a new block being added to a distributed ledger 520E, according to example embodiments, and FIG. 5D illustrates the contents of FIG. 5E's new data block structure 530E for blockchain, according to example embodiments. Referring to FIG. 5E, clients (not shown) may submit transactions to blockchain nodes 511E, 512E, and/or 513E. Clients may be instructions received from any source to enact activity on the blockchain 522E. As an example, clients may be applications that act on behalf of a requester, such as a device, person, or entity to propose transactions for the blockchain. The plurality of blockchain peers (e.g., blockchain nodes 511E, 512E, and 513E) may maintain a state of the blockchain network and a copy of the distributed ledger 520E. Different types of blockchain nodes/peers may be present in the blockchain network including endorsing peers which simulate and endorse transactions proposed by clients and committing peers which verify endorsements, validate transactions, and commit transactions to the distributed ledger 520E. In this example, the blockchain nodes 511E, 512E, and 513E may perform the role of endorser node, committer node, or both.

[0171] The distributed ledger 520E includes a blockchain which stores immutable, sequenced records in blocks, and a state database 524E (current world state) maintaining a current state of the blockchain 522E. One distributed ledger 520E may exist per channel and each peer maintains its own copy of the distributed ledger 520E for each channel of which they are a member. The blockchain 522E is a transaction log, structured as hash-linked blocks where each block contains a sequence of N transactions. The linking of the blocks (shown by arrows in FIG. 5E) may be generated by adding a hash of a prior block's header within a block header of a current block. In this way, all transactions on the blockchain 522E are sequenced and cryptographically linked together preventing tampering with blockchain data without breaking the hash links. Furthermore, because of the links, the latest block in the blockchain 522E represents every transaction that has come before it. The blockchain 522E may be stored on a peer file system (local or attached storage), which supports an append-only blockchain workload.

[0172] The current state of the blockchain 522E and the distributed ledger 520E may be stored in the state database 524E. Here, the current state data represents the latest values for all keys ever included in the chain transaction log of the blockchain 522E. Chaincode invocations execute transactions against the current state in the state database 524E. To make these chaincode interactions extremely efficient, the latest values of all keys are stored in the state database 524E. The state database 524E may include an indexed view into the transaction log of the blockchain 522E, and it can therefore be regenerated from the chain at any time. The state database 524E may automatically get recovered (or generated if needed) upon peer startup before transactions are accepted.

[0173] Endorsing nodes receive transactions from clients and endorse the transaction based on simulated results. Endorsing nodes hold smart contracts which simulate the transaction proposals. When an endorsing node endorses a transaction, the endorsing node creates a transaction endorsement which is a signed response from the endorsing node to the client application indicating the endorsement of the simulated transaction. The method of endorsing a transaction depends on an endorsement policy which may be specified within chaincode. An example of an endorsement policy is "the majority of endorsing peers must endorse the transaction." Different channels may have different endorsement policies. Endorsed transactions are forwarded by the client application to the ordering service 510E.

[0174] The ordering service 510E accepts endorsed transactions, orders them into a block, and delivers the blocks to the committing peers. For example, the ordering service 510E may initiate a new block when a threshold of transactions has been reached, a timer times out, or another condition. In the example of FIG. 5E, blockchain node 512E is a committing peer that has received a new data block 530E for storage on blockchain 522E. The first block in the blockchain may be referred to as a genesis block which includes information about the blockchain, its members, the

data stored therein, etc.

[0175] The ordering service **510E** may be made up of a cluster of orderers. The ordering service **510E** does not process transactions, smart contracts, or maintain the shared ledger. Rather, the ordering service **510E** may accept the endorsed transactions and specifies the order in which those transactions are committed to the distributed ledger **522E**. The architecture of the blockchain network may be designed such that the specific implementation of 'ordering' becomes a pluggable component.

[0176] Transactions are written to the distributed ledger **520E** in a consistent order. The order of transactions is established to ensure that the updates to the state database **524E** are valid when they are committed to the network. Unlike a cryptocurrency blockchain system where ordering occurs through the solving of a cryptographic puzzle, or mining, in this example the parties of the distributed ledger **520E** may choose the ordering mechanism that best suits the network.

[0177] When the ordering service **510E** initializes a new data block **530E**, the new data block **530E** may be broadcast to committing peers (e.g., blockchain nodes **511E**, **512E**, and **513E**). In response, each committing peer validates the transaction within the new data block **530E** by checking to make sure that the read set and the write set still match the current world state in the state database **524E**. Specifically, the committing peer can determine whether the read data that existed when the endorsers simulated the transaction is identical to the current world state in the state database **524E**. When the committing peer validates the transaction, the transaction is written to the blockchain **522E** on the distributed ledger **520E**, and the state database **524E** is updated with the write data from the read-write set. If a transaction fails, that is, if the committing peer finds that the read-write set does not match the current world state in the state database **524E**, the transaction ordered into a block will still be included in that block, but it will be marked as invalid, and the state database **524E** will not be updated.

[0178] Referring to FIG. 5F **500F**, a new data block **530** (also referred to as a data block) that is stored on the blockchain **522E** of the distributed ledger **520E** may include multiple data segments such as a block header **540**, block data **550**, and block metadata **560**. It should be appreciated that the various depicted blocks and their contents, such as new data block **530** and its contents shown in FIG. 5F, are merely examples and are not meant to limit the scope of the example embodiments. The new data block **530** may store transactional information of N transaction(s) (e.g., 1, 10, 100, 500, 1000, 2000, 3000, etc.) within the block data **550**. The new data block **530** may also include a link to a previous block (e.g., on the blockchain **522E** in FIG. 5E) within the block header **540**. In particular, the block header **540** may include a hash of a previous block's header. The block header **540** may also include a unique block number, a hash of the block data **550** of the new data block **530**, and the like. The block number of the new data block **530** may be unique and assigned in various orders, such as an incremental/sequential order starting from zero.

[0179] The block data **550** may store transactional information of each transaction that is recorded within the new data block **530**. For example, the transaction data may include one or more of a type of the transaction, a version, a timestamp, a channel ID of the distributed ledger **520E** (shown in FIG. 5E), a transaction ID, an epoch, a payload visibility, a chaincode path (deploy tx), a chaincode name, a chaincode version, input (chaincode and functions), a client (creator) identify such as a public key and certificate, a signature of the client, identities of endorsers, endorser signatures, a proposal hash, chaincode events, response status, namespace, a read set (list of key and version read by the transaction, etc.), a write set (list of key and value, etc.), a start key, an end key, a list of keys, a Merkel tree query summary, and the like. The transaction data may be stored for each of the N transactions.

[0180] In one embodiment of the instant solution, the blockchain data **563** may include data comprising one or more of determining a remaining time until charging of a vehicle is complete and transmitting content to the vehicle based on the remaining time.

[0181] Although in FIG. 5F the blockchain data **563** is depicted in the block data **550** but may also

be located in the block header **540** or the block metadata **560**.

[0182] The block metadata **560** may store multiple fields of metadata (e.g., as a byte array, etc.). Metadata fields may include signature on block creation, a reference to a last configuration block, a transaction filter identifying valid and invalid transactions within the block, last offset persisted of an ordering service that ordered the block, and the like. The signature, the last configuration block, and the orderer metadata may be added by the ordering service **510E** in FIG. 5E. Meanwhile, a committer of the block (such as blockchain node **512E** in FIG. 5E) may add validity/invalidity information based on an endorsement policy, verification of read/write sets, and the like. The transaction filter may include a byte array of a size equal to the number of transactions in the block data and a validation code identifying whether a transaction was valid/invalid.

[0183] The above embodiments may be implemented in hardware, in a computer program executed by a processor, in firmware, or in a combination of the above. A computer program may be embodied on a computer readable medium, such as a storage medium. For example, a computer program may reside in random access memory (“RAM”), flash memory, read-only memory (“ROM”), erasable programmable read-only memory (“EPROM”), electrically erasable programmable read-only memory (“EEPROM”), registers, hard disk, a removable disk, a compact disk read-only memory (“CD-ROM”), or any other form of storage medium known in the art.

[0184] An exemplary storage medium may be coupled to the processor such that the processor may read information from, and write information to, the storage medium. In the alternative, the storage medium may be integral to the processor. The processor and the storage medium may reside in an application-specific integrated circuit (“ASIC”). In the alternative, the processor and the storage medium may reside as discrete components. For example, FIG. 6 illustrates an example computer system architecture **600**, which may represent or be integrated in any of the above-described components, etc.

[0185] FIG. 6 illustrates a computing environment according to example embodiments. FIG. 6 is not intended to suggest any limitation as to the scope of use or functionality of embodiments of the application described herein. Regardless, the computing environment **600** can be implemented to perform any of the functionalities described herein. In computer environment **600**, computing system **601** is operational within numerous other general-purpose or special-purpose computing system environments or configurations.

[0186] Computing system **601** may take the form of a desktop computer, laptop computer, tablet computer, smartphone, smartwatch or other wearable computer, server computer system, thin client, thick client, network PC, minicomputer system, mainframe computer, quantum computer, and distributed cloud computing environment that include any of the described systems or devices, and the like or any other form of computer or mobile device now known or to be developed in the future that is capable of running a program, accessing a network **650** or querying a database. Depending upon the technology, the performance of a computer-implemented method may be distributed among multiple computers and between multiple locations. However, in this presentation of the computing environment **600**, a detailed discussion is focused on a single computer, specifically computing system **601**, to keep the presentation as simple as possible.

[0187] Computing system **601** may be located in a cloud, even though it is not shown in a cloud in FIG. 6. On the other hand, computing system **601** is not required to be in a cloud except to any extent as may be affirmatively indicated. Computing system **601** may be described in the general context of computer system-executable instructions, such as program modules, executed by a computing system **601**. Generally, program modules may include routines, programs, objects, components, logic, data structures, and so on that perform tasks or implement certain abstract data types. As shown in FIG. 6, computing system **601** in computing environment **600** is shown in the form of a general-purpose computing device. The components of computing system **601** may include but are not limited to, one or more processors or processing units **602**, a system memory **630**, and a bus **620** that couples various system components, including system memory **630** to

processing unit **602**.

[0188] Processing unit **602** includes one or more computer processors of any type now known or to be developed. The processing unit **602** may contain circuitry distributed over multiple integrated circuit chips. The processing unit **602** may also implement multiple processor threads and multiple processor cores. Cache **632** is a memory that may be in the processor chip package(s) or located “off-chip,” as depicted in FIG. 6. Cache **632** is typically used for data or code that the threads or cores running on the processing unit **602** should be available for rapid access. In some computing environments, processing unit **602** may be designed to work with qubits and perform quantum computing.

[0189] Network adapter **603** enables the computing system **601** to connect and communicate with one or more networks **650**, such as a local area network (LAN), a wide area network (WAN), and/or a public network (e.g., the Internet). It bridges the computer's internal bus **620** and the external network, exchanging data efficiently and reliably. The network adapter **603** may include hardware, such as modems or Wi-Fi signal transceivers, and software for packetizing and/or de-packetizing data for communication network transmission. Network adapter **603** supports various communication protocols to ensure compatibility with network standards. For Ethernet connections, it adheres to protocols such as IEEE 802.3, while for wireless communications, it might support IEEE 802.11 standards, Bluetooth, near-field communication (NFC), or other network wireless radio standards.

[0190] Computing system **601** may include a removable/non-removable, volatile/non-volatile computer storage device **610**. By way of example only, storage device **610** can be a non-removable, non-volatile magnetic media (not shown and typically called a “hard drive”). One or more data interfaces can connect it to the bus **620**. In embodiments where computing system **601** is required to have a large amount of storage (for example, where computing system **601** locally stores and manages a large database), then this storage may be provided by storage devices **610** designed for storing very large amounts of data, such as a storage area network (SAN) that is shared by multiple, geographically distributed computers.

[0191] The operating system **611** is software that manages computing system **601** hardware resources and provides common services for computer programs. Operating system **611** may take several forms, such as various known proprietary operating systems or open-source Portable Operating System Interface type operating systems that employ a kernel.

[0192] The bus **620** represents one or more of several types of bus structures, including a memory bus or memory controller, a peripheral bus, an accelerated graphics port, and a processor or local bus using various bus architectures. By way of example, and not limitation, such architectures include Industry Standard Architecture (ISA) buses, Micro Channel Architecture (MCA) buses, Enhanced ISA (EISA) buses, Video Electronics Standards Association (VESA) local buses, and Peripheral Component Interconnect (PCI) bus. The bus **620** is the signal conduction paths that allow the various components of computing system **601** to communicate with each other.

[0193] Memory **630** is any volatile memory now known or to be developed in the future. Examples include dynamic random-access memory (RAM **631**) or static type RAM **631**. Typically, the volatile memory is characterized by random access, but this is not required unless affirmatively indicated. In computing system **601**, memory **630** is in a single package and is internal to computing system **601**, but alternatively or additionally, the volatile memory may be distributed over multiple packages and/or located externally with respect to computing system **601**. By way of example only, memory **630** can be provided for reading from and writing to a non-removable, non-volatile magnetic media (shown as storage device **610**, and typically called a “hard drive”). Memory **630** may include at least one program product having a set (e.g., at least one) of program modules configured to carry out various functions. A typical computing system **601** may include cache **632**, a specialized volatile memory generally faster than RAM **631** and generally located closer to the processing unit **602**. Cache **632** stores frequently accessed data and instructions

accessed by the processing unit **602** to speed up processing time. The computing system **601** may include non-volatile memory **633** in ROM, PROM, EEPROM, and flash memory. Non-volatile memory **633** often contains programming instructions for starting the computer, including the basic input/output system (BIOS) and information required to start the operating system **611**.

[0194] Computing system **601** may also communicate with one or more peripheral devices **641** via an input/output (I/O) interface **640**. Such devices may include a keyboard, a pointing device, a display, etc.; one or more devices that enable a user to interact with computing system **601**; and/or any devices (e.g., network card, modem, etc.) that enable computing system **601** to communicate with one or more other computing devices. Such communication can occur via I/O interfaces **640**. As depicted, I/O interface **640** communicates with the other components of computing system **601** via bus **620**.

[0195] Network **650** is any computer network that can receive and/or transmit data. Network **650** can include a WAN, LAN, private cloud, or public Internet, capable of communicating computer data over non-local distances by any technology that is now known or to be developed in the future. Any connection depicted can be wired and/or wireless and may traverse other components that are not shown. In some embodiments, a network **650** may be replaced and/or supplemented by LANs designed to communicate data between devices located in a local area, such as a Wi-Fi network. The network **650** typically includes computer hardware such as copper transmission cables, optical transmission fibers, wireless transmission, routers, firewalls, switches, gateway computers, and edge servers, and network infrastructure known now or to be developed in the future. Computing system **601** connects to network **650** via network adapter **603** and bus **620**.

[0196] User devices **651** are any computer systems used and controlled by an end user in connection with computing system **601**. For example, in a hypothetical case where computing system **601** is designed to provide a recommendation to an end user, this recommendation may typically be communicated from network adapter **603** of computing system **601** through network **650** to a user device **651**, allowing user device **651** to display, or otherwise present, the recommendation to an end user. User devices can be a wide array of devices, including personal computers (PCs), laptops, tablet, hand-held, mobile phones, etc.

[0197] Remote servers **660** are any computers that serve at least some data and/or functionality over a network **650**, for example, WAN, a virtual private network (VPN), a private cloud, or via the Internet to computing system **601**. These networks **650** may communicate with a LAN to reach users. The user interface may include a web browser or an application that facilitates communication between the user and remote data. Such applications have been called “thin” desktops or “thin clients.” Thin clients typically incorporate software programs to emulate desktop sessions. Mobile applications can also be used. Remote servers **660** can also host remote databases **661**, with the database located on one remote server **660** or distributed across multiple remote servers **660**. Remote databases **661** are accessible from database client applications installed locally on the remote server **660**, other remote servers **660**, user devices **651**, or computing system **601** across a network **650**.

[0198] A public cloud **670** is an on-demand availability of computer system resources, including data storage and computing power, without direct active management by the user. Public clouds **670** are often distributed, with data centers in multiple locations for availability and performance. Computing resources on public clouds (**670**) are shared across multiple tenants through virtual computing environments comprising virtual machines **671**, databases **672**, containers **673**, and other resources. A container **673** is an isolated, lightweight software for running an application on the host operating system **611**. Containers **673** are built on top of the host operating system's kernel and contain only applications and some lightweight operating system APIs and services. In contrast, virtual machine **671** is a software layer that includes a complete operating system **611** and kernel. Virtual machines **671** are built on top of a hypervisor emulation layer designed to abstract a host computer's hardware from the operating software environment. Public clouds **670** generally

offer hosted databases 672 abstracting high-level database management activities. It should be further understood that one or more of the elements described or depicted in FIG. 6 can perform one or more of the actions, functionalities, or features described or depicted herein.

[0199] Although an exemplary embodiment of at least one of a system, method, and non-transitory computer readable medium has been illustrated in the accompanied drawings and described in the foregoing detailed description, it will be understood that the application is not limited to the embodiments disclosed, but is capable of numerous rearrangements, modifications, and substitutions as set forth and defined by the following claims. For example, the capabilities of the system of the various figures can be performed by one or more of the modules or components described herein or in a distributed architecture and may include a transmitter, receiver, or pair of both. For example, all or part of the functionality performed by the individual modules, may be performed by one or more of these modules. Further, the functionality described herein may be performed at various times and in relation to various events, internal or external to the modules or components. Also, the information sent between various modules can be sent between the modules via at least one of: a data network, the Internet, a voice network, an Internet Protocol network, a wireless device, a wired device and/or via plurality of protocols. Also, the messages sent or received by any of the modules may be sent or received directly and/or via one or more of the other modules.

[0200] One skilled in the art will appreciate that a “system” may be embodied as a personal computer, a server, a console, a personal digital assistant (PDA), a cell phone, a tablet computing device, a smartphone or any other suitable computing device, or combination of devices. Presenting the above-described functions as being performed by a “system” is not intended to limit the scope of the present application in any way but is intended to provide one example of many embodiments. Indeed, methods, systems and apparatuses disclosed herein may be implemented in localized and distributed forms consistent with computing technology.

[0201] It should be noted that some of the system features described in this specification have been presented as modules to more particularly emphasize their implementation independence. For example, a module may be implemented as a hardware circuit comprising custom very-large-scale integration (VLSI) circuits or gate arrays, off-the-shelf semiconductors such as logic chips, transistors, or other discrete components. A module may also be implemented in programmable hardware devices such as field-programmable gate arrays, programmable array logic, programmable logic devices, graphics processing units, or the like.

[0202] A module may also be at least partially implemented in software for execution by various types of processors. An identified unit of executable code may, for instance, comprise one or more physical or logical blocks of computer instructions that may, for instance, be organized as an object, procedure, or function. Nevertheless, the executables of an identified module need not be physically located together but may comprise disparate instructions stored in different locations that, when joined logically together, comprise the module and achieve the stated purpose for the module. Further, modules may be stored on a computer-readable medium, which may be, for instance, a hard disk drive, flash device, random access memory (RAM), tape, or any other such medium used to store data.

[0203] Indeed, a module of executable code may be a single instruction, or many instructions, and may even be distributed over several different code segments, among different programs, and across several memory devices. Similarly, operational data may be identified and illustrated herein within modules and may be embodied in any suitable form and organized within any suitable type of data structure. The operational data may be collected as a single data set or may be distributed over different locations, including over different storage devices, and may exist, at least partially, merely as electronic signals on a system or network.

[0204] It will be readily understood that the components of the application, as generally described and illustrated in the figures herein, may be arranged and designed in a wide variety of different

configurations. Thus, the detailed description of the embodiments is not intended to limit the scope of the application as claimed but is merely representative of selected embodiments of the application.

[0205] One having ordinary skill in the art will readily understand that the above may be practiced with steps in a different order and/or with hardware elements in configurations that are different from those which are disclosed. Therefore, although the application has been described based upon these preferred embodiments, it would be apparent to those of skill in the art that certain modifications, variations, and alternative constructions would be apparent.

[0206] While preferred embodiments of the present application have been described, it is to be understood that the embodiments described are illustrative only and the scope of the application is to be defined solely by the appended claims when considered with a full range of equivalents and modifications (e.g., protocols, hardware devices, software platforms etc.) thereto.

Claims

1. A method, comprising: determining a remaining time until charging of a vehicle is complete; and transmitting content to the vehicle based on the remaining time.
2. The method of claim 1, comprising: identifying an application associated with a profile of an occupant of the vehicle; and initiating the application while the vehicle is charging when the remaining time is within a time duration to use the application.
3. The method of claim 1, comprising: identifying one or more actions performed by an occupant while using an application; and enabling one or more vehicle functions responsive to the one or more actions identified in the application.
4. The method of claim 3, wherein the one or more vehicle functions comprise a chair movement, a change in air ventilation speed or temperature, a change in light control, and a change in an air infuser.
5. The method of claim 1, wherein the content transmitted is streaming content of a game associated with an occupant of the vehicle.
6. The method of claim 2, comprising determining a game session associated with the application has ended prematurely at an updated remaining time; and determining not to initiate another game session based on the updated remaining time.
7. The method of claim 2, comprising determining the charging is complete; determining a game session associated with the application is not completed; and forwarding game session data to a user device.
8. A system comprising a memory and a processor configured to: determine a remaining time until charging of a vehicle is complete; and transmit content to the vehicle based on the remaining time.
9. The system of claim 8, wherein the processor is further configured to: identify an application associated with a profile of an occupant of the vehicle; and initiate the application while the vehicle is charging when the remaining time is within a time duration to use the application.
10. The system of claim 8, wherein the processor is further configured to: identify one or more actions performed by an occupant while using an application; and enable one or more vehicle functions responsive to the one or more actions identified in the application.
11. The system of claim 10, wherein the one or more vehicle functions comprise a chair movement, a change in air ventilation speed or temperature, a change in light control, and a change in an air infuser.
12. The system of claim 8, wherein the content transmitted is streaming content of a game associated with an occupant of the vehicle.
13. The system of claim 9, wherein the processor is further configured to: determine a game session associated with the application has ended prematurely at an updated remaining time; and determine not to initiate another game session based on the updated remaining time.

- 14.** The system of claim 9, wherein the processor is further configured to: determine the charging is complete; determine a game session associated with the application is not completed; and forwarding game session data to a user device.
- 15.** A non-transitory computer readable medium configured to store instructions that when executed cause a processor to perform: determining a remaining time until charging of a vehicle is complete; and transmitting content to the vehicle based on the remaining time.
- 16.** The non-transitory computer readable medium of claim 15, wherein the processor is further configured to perform: identifying an application associated with a profile of an occupant of the vehicle; and initiating the application while the vehicle is charging when the remaining time is within a time duration to use the application.
- 17.** The non-transitory computer readable medium of claim 15, wherein the processor is further configured to perform: identifying one or more actions performed by an occupant while using an application; and enabling one or more vehicle functions responsive to the one or more actions identified in the application.
- 18.** The non-transitory computer readable medium of claim 17, wherein the one or more vehicle functions comprise a chair movement, a change in air ventilation speed or temperature, a change in light control, and a change in an air infuser.
- 19.** The non-transitory computer readable medium of claim 15, wherein the content transmitted is streaming content of a game associated with an occupant of the vehicle.
- 20.** The non-transitory computer readable medium of claim 16, wherein the processor is further configured to perform: determining a game session associated with the application has ended prematurely at an updated remaining time; and determining not to initiate another game session based on the updated remaining time.
-